

EDUC 130 Final Research Paper

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Effects of Group Dynamics on Student Conjectures

Abstract

When teachers assign group work, students are expected to work together on a provided problem. What teachers may not notice is how they participate and collaborate, and how they affect a group's progress and understanding. The paper investigates how students collaborate and share ideas for finding patterns and transformations of generalized magic squares by using the basic 3-by-3 magic square as the base solution. The study uses student-used digital interfaces like Google Slides and Google Drawings, video recordings, and multimodal student responses to conjecture about the effect of group discussion and participation, or the lack thereof, on student reasoning and conjecture. The study's results demonstrate a noticeable difference in active cooperation and co-construction regarding students' reasoning, favoring co-construction and both the sharing and revision of ideas over mathematical maturity. These findings reinforce the findings of Barron (2003); both conceptual and social cooperation are necessary to tackle a difficult and wide-ranging problem regardless of mathematical maturity.

Introduction

Teachers often use group work not only as a way to build students' collaboration skills, but also as an efficient, student-oriented learning tool (White et al., 2007) that helps students better understand and explain concepts (Allen, 2012). However, group work often fails to achieve these benefits because of unequal collaboration (White et al., 2007), disjoint collaboration between more individualistic students (Allen, 2012), and other personal or cognitive influences, all of which can lead to situations like "skipconnecting," which Barron (2003) describes as a speaker "declines to talk about the previous speaker's topic and reasserts his own." In Education 130, several solutions have been presented to address these issues, most notably, assigning specific roles (Allen, 2012). Notably, according to the course reading studies conducted by Barron (2003), student engagement and understanding were noticeably better when students openly collaborated, especially when their collaboration involved the co-construction of solutions – the building of a complete idea based on the combined ideas and information provided by students. In line with Education 130's course objectives, I wanted to see whether students would naturally ensure their group's success, and learn directly

how students consider, collaborate, and solve complex mathematical problems.

Purpose of the Study

I conducted the study with the idea of keeping not only Barron's terminology in mind when observing students, in this case, the undergraduate students I could make available, but also to see whether students would spontaneously form cohesive and productive groups without intervention. As a sufficient challenge to the undergraduate participants, I used the magic square problem as a basis for introducing vector spaces, a mathematical topic typically left for college mathematics. I used a challenging generalization of an already hard problem because of the numerically minimal patterns needed to introduce vector space axioms, and that axioms are considered true by definition, thus not requiring fundamental proofs. Rather, the issue of discovering new magic squares, connecting them to patterns, and generalizing such patterns is a multifaceted construction process that demanded student cooperation, reasoning, and conjecturing in a manner that establishes what Ellis (2007) establishes as a "bidirectional" connection "between generalization and [algebraic] justification" – a relevant facet of the research that is used sparingly but strongly. To understand how group work dynamics facilitate the generation of generalizations from algebraic reasoning, this study considers: How do undergraduate participants take up, interpret, and revise one another's conjectures about transformations that preserve a 3-by-3 magic square during small-group discussion, and how does this process differ among different groups?

A Research Introduction to Magic Squares as a Vector Space

The magic square problem in mathematics involves placing the first n^2 numbers in an n -by- n grid, so that the sum of each row, column, and diagonal is the same. Despite having a low technical prerequisite – i.e., basic arithmetic – it has uses in various fields of mathematics, ranging from Algebra and Geometry and Linear Algebra and Number Theory. This unique property and usage of the magic square make it exceptionally rich, while also being deceptively difficult at base. For this study, I focus on generalized 3-by-3 magic squares, using the normal 3-by-3 magic square, otherwise known as the Lo Shu square, as a basis (pun not intended) for sparking student experimentation with the square's properties and transformation patterns in pursuit of finding other magic squares without the restrictions of using the numbers 1 to 9 or avoiding repeats. Removing these restrictions enables students to discover a 3-dimensional vector space that cannot be easily understood or represented without active collaboration and co-construction. It also

allows students to make conjectures about its properties, i.e., the 10 vector space axioms, without knowing about it in the process. Even if students' ideas seem to vary wildly, their ideas, put together, may generate a usable pattern or describable axiom.

I was inspired to take a linear algebra approach to magic squares based on the works of Lancaster (1979), who analyzed magic squares and their inverses, and Eperson (1959), who gave a general formula for generating 3-by-3 magic squares using three variables. This led me to consider magic squares as a vector space, as all 10 axioms of vector spaces apply to the vector space of 3-by-3 magic squares. While I came up with this idea independently of any sources, Sawada (1974) provided much more detailed insight for a proper motivation of magic squares as a vector space. As such, I focused my study on whether students could find patterns – namely, certain transformations that relate to vector space axioms – and other interesting properties of magic squares. Unlike the lesson, it does not evaluate student understanding of a vector space, which will be discussed later. Given the difficulty of such an endeavor, group work was necessary, but it provided a useful dimension to analyze student reasoning and collaboration. Further information about the magic square is available in the Problem Analysis.

Design

Problem Analysis of the Magic Square

Introduction

The magic square problem in mathematics is an interesting historical problem originating in ancient China and India (Andrews, 1960), which involves arranging the first n^2 numbers in an n -by- n grid such that each row, column, and diagonal has the same sum. As a “technically” simple problem – the only prerequisite is knowing how to add numbers – the magic squares problem is highly accessible, such that everyone in grade 1 and above can hypothetically solve this. This high versatility allows for its use in a variety of mathematical concepts and CA Common Core State Standards for Mathematics (CCSSM) learning goals, including but not limited to: “Add and subtract within 20” (Grades 1 and 2), “Generate and analyze patterns”, “Recognize a line of symmetry” (Grade 4), “Solve systems of equations” (Algebra I, as well as Linear Algebra) (California Department of Education, 2013), probability, counting, basic statistics, and

discrete mathematics. Thus, not only can this problem be taught in various contexts, but it can also be questioned in even more ways, spanning all facets of mathematical education across all of K-12 and even college. As a mostly mathematical problem in itself, I personally wanted to use this problem, as I would have the most expertise to solve, understand, and elaborate the concepts that allow the solving of the square. Not only is it a hard problem, which makes it engaging on its own, but the ways that the problem can branch off into are even more engaging, delving both into theoretical and technical applications of the problem that promote creative problem solving, strategy-making, and diversity in mathematical thinking, all of which have uses of varying importance not only in higher-level mathematics but also in general.

The Magic Square Problem

Solving the magic square problem is not necessarily a simple task. As a simple counting example, for a 3-by-3 square, we have 362880, or $9!$ (factorial) different placements of the numbers 1 through 9, which originates from arbitrarily placing 1 through 9 in that order, choosing from 9 squares, 8 squares, $\dots$, and 1 square. To guess the solution of the 3x3 version, there exists only one unique solution, which the Chinese called the Lo Shu square (Andrews, 1960), with the other seven solutions being symmetries or rotations of the Lo Shu square. This, in itself, is a puzzle of not only recognizing these symmetries and rotations, but also of recognizing the number of axes of symmetry the square has, of which there are four. If one randomly guesses the square, one has an $8/362880$ chance of stumbling upon the answer, or approximately 0.0022%. In essence, to solve the magic square efficiently, students must hypothesize and develop potential solving strategies or patterns and test them in the square itself. The numerous ways to approach, think, and solve the 3-by-3 magic square, let alone a general n -by- n , make this problem especially rich.

4	9	2
3	5	7
8	1	6

Figure 1: The Lo Shu square

Solving the Problem

When we began solving the problem individually, I understood that each row, column, and diagonal would have the same sum. I decided that to figure out what that sum is, I should first add up the numbers 1 to 9. I then used my mathematical knowledge that the sum of the first n numbers is $\frac{n(n+1)}{2}$; I found that the sum of the numbers from 1 to 9 is $\frac{9 \cdot 10}{2} = 45$. I used my understanding that each of the 3 rows and 3 columns would each use 3 numbers that sum to the same magic sum, so I divided 45 by 3 to get 15. That led me to recall that the magic sum of an n -by- n magic square was $\frac{n(n^2 + 1)}{2}$, which I had looked up several weeks earlier when I was trying to learn MATLAB for MATH W128A. After finding this magic sum, my initial thought was to write out all the triplets whose sum was 15. However, because I was not willing to do that, I decided to be lazy and place large numbers on the edges. In the moment, I thought this idea could work, as the numbers on the edge would contribute to fewer sums (just two) than other spots (three or four). However, I quickly came to a roadblock when I tried placing the remaining numbers, as I kept getting two rows or columns that did not add to 15. After that, I decided not to be lazy and actually try to solve it. I began writing down each triplet whose sum was 15, and noticed that while I was correct that 9 should be on an edge, as it could only be used with 1 and 5, and 2 and 4, 8 should have been in the corner, since it is used with 1 and 6, 2 and 5, and 3 and 4. Proceeding with my triplet writing, I quickly deduced where 7 went, then 6, then 5, and I placed the rest of the numbers easily.

During group work on the problem, or for my group, mostly discussion, the other members of my group also solved the square during our free time. I presented my mathematical approach to the problem, initially just with formulas, which my other group members had trouble understanding. Thus, I spent some time explaining why the formulas worked, explaining that these sums of the first n numbers were called triangular numbers, and elaborated on how I translated my reasoning in deriving the magic sum into mathematical notation. After that, I did learn some methods from my other group members that were definitely a new way, at least for me, to think about the problem that I would not have tried or thought about explicitly. The other members realized that 5 belongs in the middle, especially after one of them tried putting other numbers in the middle square to no avail. The other group member tried placing the other eight numbers at random, but he soon realized that the odd numbers should also be on the edges, such that each row, column, and diagonal has an odd number of odd integers. Using number theory, since the magic sum is odd, one can only achieve an odd number if we have an odd number of odd numbers. If we have an even number of odd

numbers, then the sum must be even based on divisibility rules. One of my thoughts as we were discussing is whether the magic square is unique for any magic square, as when I was looking at the magic() function in MATLAB, I saw that it only outputted the same magic square. From that, I thought that only one unique magic square exists, but when I looked it up several seconds later, I realized I was wrong for larger magic squares. What I did find interesting was that the 3-by-3 was, truly, unique, apart from symmetry and rotation. Once we returned to full-class discussion, the other students approached the problem in one of several ways (or a combination of them):

1. Using mathematical expressions for the total sum or magic sum. (Part of my successful approach)
2. Placing 5 in the middle and proceeding from there.
3. Recognizing that each row, column, and diagonal must have an odd number of odd numbers.
4. Writing all triplets that add to the magic sum. (My successful approach)
5. Placing the larger numbers on the edges or corners. (My initial approach)
6. The Siamese Method

More methods or approaches to solve the magic square exist, but Education 130 students mentioned these methods. Overall, when approaching the magic squares problem, basic arithmetic and some degree of pattern recognition (such as pairing numbers that add to 10, or recognizing that the magic sum is 15) are enough to attempt the problem with a reasonable probability of success.

Using Mathematical Expressions for the Total Sum or Magic Sum

The sum of all the numbers and the magic sum for an n -by- n magic square, as stated in "Solving the Problem", are $\frac{n^2(n^2 + 1)}{2}$ and $\frac{n(n^2 + 1)}{2}$, respectively. Students getting to here will have established structure and completed a sub-goal in the process. If students find the magic sum, but are unsure how to continue, I would remind them that all rows, columns, and diagonals have the same sum, and suggest testing certain number placements, while keeping the magic sum in mind. Students who are more familiar with Algebra 2 content or discrete mathematics will likely take this approach first before trying anything else.

Placing 5 in the Middle

Many others recognized that, since 5 is the mean of the numbers and 4 pairs using the remaining eight numbers sum to 10, 5 should be placed in the middle. Since one can only use 5 once, and the rules prevent using another 5 that would make 10, it must be in the middle. The more generalized reasons why this works will be explained in a later section. When I heard classmates mention these strategies, I realized that one does not require much mathematical knowledge; a good amount of conjecturing, testing, and using common sense, along with just some math, is enough for over half of everyone to solve the problem. This method uses constraints, tests cases, and uses symmetry in some form or another. If students are unsure how to proceed, I would suggest that they try test cases with number placements and see what the implications are based on the result of those placements. Students familiar with symmetry in their lives will likely find this method quicker than other methods.

Recognizing that each row, column, and diagonal must have an odd number of odd numbers

As explained above, we can only achieve an odd magic sum with an odd number of odd numbers. As a creative find of the problem's structure, future progress on the problem should include actually using that structure to create some test cases, along with constraining certain odd numbers to specific placements.

Writing All Triplets that Add to the Magic Sum

While this strategy is certainly slow, requiring you to write out the 8 combinations of 3 numbers (or 24 permutations), the number of times each number appears becomes a very powerful indicator as to where each number should be placed. If 5 appears four times, it must go in the center (for a 3-by-3 square). If a number appears twice, it should go on the edge. If a number appears three times, it should go in the corner. This strategy works more generally by writing all n -tuples for an n -by- n magic square, but it is more cumbersome for larger squares. This was the strategy that helped me find the 3-by-3 magic square quite efficiently, and not many others approached it in this way. While it may have been okay for this smaller square, the method would be highly inefficient for larger squares. Regardless, this makes efficient use of the problem's structure, and only requires testing cases or working backwards from any given row, column, or diagonal.

Placing the larger numbers on the edges or corners

While this method is partially flawed, it involves testing many cases, which helps eliminate some of the remaining cases. However, despite the intuition behind this idea, it only partially constrains the problem, but not noticeably. In reality, it would be better to nudge a student trying this idea to think of another strategy.

The Siamese Method

One student mentioned this method, which works for all odd-length magic squares. Starting with 1 in the top-center entry, one continually places the next number diagonally to the top-right, wrapping from the top to the bottom and from the right to the left. If the square is occupied, one should place the next number below the current number instead. One repeats the process until all numbers are placed. However, a student will likely not naturally use this method unless he or she has already been exposed to the problem and method beforehand. A similar, but potentially more familiar alternative, exists within a group of such methods, as discussed in the next section.

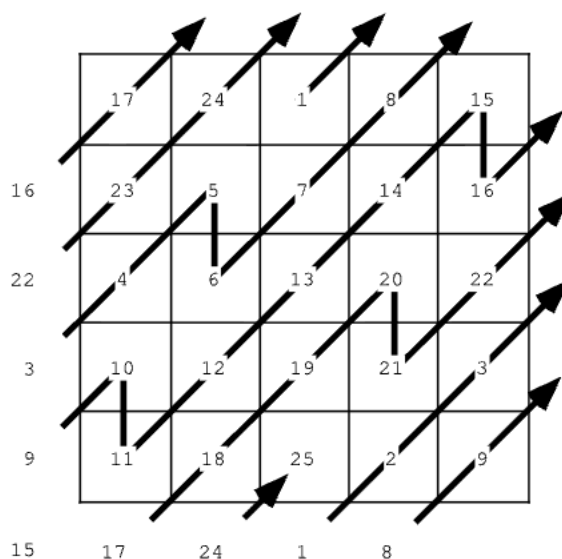


Figure 2: Siamese Method. Image taken from Weisstein.

Other Solutions

Andrews (1960) compiled several methods and mathematical analyses for solving the magic square. One method uses knight jumps. It is effectively similar to the Siamese Method in steps and uses, but this method uses knight

moves in any initial direction instead of diagonal moves, as long as it is consistent. A generalization of both the knight jump method and the Siamese method is what Weisstein (n.d.) conceptualizes as a combination of an “ordinary” and a “break” vector. The Siamese method consists of the first combination of an ordinary and a break vector.

ordinary vector	break vector	sumdiffs	magic squares	panmagic squares
(1, − 1)	(0, 1)	(1, 3)	$2k + 1$	none
(1, − 1)	(0, 2)	(0, 2)	$6k \pm 1$	none
(2, 1)	(1, − 2)	(1, 2, 3, 4)	$6k \pm 1$	none
(2, 1)	(1, − 1)	(0, 1, 2, 3)	$6k \pm 1$	$6k \pm 1$
(2, 1)	(1, 0)	(0, 1, 2)	$2k + 1$	none
(2, 1)	(1, 2)	(0, 1, 2, 3)	$6k \pm 1$	none

Figure 3: Weisstein (n.d.) Table of “Ordinary Vectors” and “Break Vectors”. Vectors assume that the square is, mathematically, a matrix. “Magic Squares” denotes the sizes for which the given combination works. “Pan-magic Squares” are squares where all off-diagonals also sum to the magic sum.

The proofs for these sets of methods rest in the use of Latin squares, and, by extension, de la Hire’s method, which we will not elaborate on (Andrews, 1960). Latin squares of size n -by- n use each of the numbers 1 to n a total of n times. If one places these numbers to create a Latin square A , and then creates an orthogonal Latin square B by rotating or flipping $(n - 1)B$ such that all pairs of (a,b) for $A_{i,j} = a, B_{i,j} = b$, then the normal n -by- n magic square is created. By placing each number in M in the same direction as the “ordinary vector” and placing each subsequent number a “break vector” away, one heuristically creates the patterns generated by the ordinary-break vector combination. This proof has roots in number theory, and could be used with this method derivation in mind.

Weisstein also notes an elementary method for solving n -by- n magic squares for n divisible by 4. Starting with a square with the numbers 1 to n^2 written in order like a number table, subdivide the square into 4-by-4 sub-squares and draw its diagonals. Find the complement sum, which is $n^2 + 1$, and for each i on a diagonal, replace it with $n^2 + 1 - i$. This method works in a mathematical sense: in each 4-by-4 square, before applying the method, each row and column had a pair of outer numbers whose sum equaled the inner pair’s sum. For example, $5 + 8$ (outer sum) = $13 = 6 + 7$ (inner sum). Let us denote this sum as S . By applying the method, we replace either both inner numbers or both outer numbers, depending on their placement, with $n^2 - 1 - i$. Taking the same example, if 5 and 8 were on the diagonal of

the small 4-by-4, then $5+8 = (n^2 + 1 - 5) + (n^2 + 1 - 8) = 2(n^2 + 1) - S$. This new sum, combined with the remaining pair, gives us a row or column sum of $S + 2(n^2 + 1) - S = 2(n^2 + 1)$. Observe that for an n -by- n magic square, each sub-square has $n/4$ squares in a single row or column. Thus, we can do $n/4 \cdot 2(n^2 + 1) = \frac{n(n^2 + 1)}{2}$, and again, obtain the magic sum for the n -by- n magic square. For the main diagonals, observe that opposite entries on the diagonals add to $n^2 + 1$; thus, for the n entries on the diagonal, we get $n/2$ sums of $n^2 + 1$, which, multiplied together, equal the magic sum.

Other perfectly sound methods exist, such as the LUX method Weisstein (n.d.) for n -by- n squares, where n is of the form $4m + 2$; however, this method will not be elaborated on.

A More Intuitive Understanding of the Magic Square

While all the concepts I have mentioned so far are mathematically dense, the problem can still be viewed through intuitive, common-sense thinking. The often-stated method of using the middle number, 5, in the center can be attributed to the general intuition of placing something not so extreme in the center. Concurrently, some symmetry searching can lead students to alternate odd and even numbers, or large and small numbers, in the outer eight boxes. Viewing the corners, edges, and center in terms of their contributions to different numbers of sums also uses pattern recognition. For chess players, the knight method (or the Siamese method, which they may attribute to a bishop) could serve as an intuitive starting point for solving the problem.

Initial Presentation of the Magic Square

I will split this section into three parts with possible follow-ups: Elementary-level (focusing on arithmetic, basic symmetry), Secondary-level (systems of equations, statistics and probability), and Advanced (advanced counting and probability, linear algebra, and number theory). The latter will be discussed later with respect to the secondary problems after completion of the 3-by-3 magic square, as well as other elementary and secondary-level uses of those problems. I intend to use the advanced-level topics with my test group, given that most of them have already graduated high school or are current undergraduates.

An Elementary Introduction to the Magic Square

This section will be split by grade and uses several learning goals and objectives from the California Common Core State Standards for Mathematics (California Department of Education, 2013).

Grade 1 is the earliest that the magic square has reasonable use in learning mathematics. The problem can be presented in a partially complete manner, with a row/column/diagonal and an additional square filled in, and the necessary statement that all of them have the same sum, with the expressed learning goals being "Understand the meaning of the equal sign, and determine if equations involving addition and subtraction are true or false" and "Determine the unknown whole number in an addition or subtraction equation relating three whole numbers". Of course, it would be up to the students to find the numbers needed to fill in the magic square; this allows students to use their intuition and hands-on knowledge from kindergarten to solve it. The problem can be similarly presented to Grade 2 under the learning goal, "Represent and solve problems involving addition and subtraction", and to Grade 3 under the learning objective, "Identify arithmetic patterns ... and explain them using properties of operations", with emphasis on $\text{odd} + \text{odd} = \text{even}$ and $\text{odd} + \text{even} = \text{odd}$. For Grades 1 and 2, physical blocks (or Lego bricks) could be used as a cheap, hands-on version of the problem for multimodal learning. This substitution enables students' tactile recall or familiarity with whatever their previous teachers did with blocks. Students here could work individually or in groups.

Some questions that may arise from these students would likely be simple questions like, "How am I supposed to know what to put here?" or "I don't have enough blocks to finish this row/column/diagonal." For example, a student might currently be stuck with a 6-5-4 diagonal and a 4 and 9 in a row:

$$\begin{bmatrix} 4 & 9 & ? \\ - & 5 & - \\ - & - & 6 \end{bmatrix}$$

. One response could be, "Well, what things are the same sum (or amount if working with blocks)?" If they identify that $4 + 5 + 6 = 4 + 9 + \square$ (or a similar identification without the 4 on both sides), then ask, "What could go there to make that true?" If they don't identify a correct equation, then ask, "Think about what this row/column/diagonal shares with something else." If a student made a mistake and is struggling to finish the problem, ask, "Is there somewhere

where you did not need some blocks?”

For Grade 4, the magic square can be presented as previously noted above, but after they solve it, present one of the two following, depending on the learning objective: For the learning objective “Understand addition and subtraction of fractions as joining and separating parts referring to the same whole”, replace the numbers 1 to 9 with, for example, $\frac{1}{4}$, $\frac{1}{2}$, ..., $\frac{9}{4}$ ”, which has the Grade 3 prerequisite that they can “express whole numbers as fractions” and “Recognize and generate simple equivalent fractions”; or, for the learning objective “Recognize a line of symmetry for a two-dimensional figure...”, change the initial given placement of the numbers. For example, if the students were initially given

$$\begin{bmatrix} 4 & 9 & - \\ - & 5 & - \\ - & - & 6 \end{bmatrix}, \quad \text{then give them} \quad \begin{bmatrix} 4 & - & - \\ 9 & 5 & - \\ - & - & 6 \end{bmatrix}$$

to hint at (one of) the square’s diagonal symmetry. Additional questions the students may ask may include, “How do I solve this?”, which is a fair question when students are not familiar at all with such a change. A straightforward response would be “Think about some similarities with the first one”.

Secondary Level Introduction

At this point, we can present the magic square as it is, empty and given only the numbers 1 to 9, and that all rows, columns, and diagonals have the same sum. I will address potential uses related to CCSSM learning goals and objectives here, and then proceed with potential questions in the next section.

For Grade 8, Algebra I, and Mathematics I, each one shares approximately the same learning goal, “Analyze and solve linear equations and pairs of simultaneous linear equations.” Since the problem can be viewed as a set of linear equations, the magic square serves to give an implicit introduction to the concept itself.

As stated in the previous subsection, the magic square can be used to demonstrate symmetry, but it can also demonstrate rotational symmetry in a more formal geometry class context. In this case, take four numbers from the student’s version of the completed magic square, and rotate it. In a statistical and probabilistic sense, the time students took to complete the magic square, the number of hints given, etc., could all be used to introduce various statistical concepts such as the mean, median, standard deviation, range, bar graphs, and linear regression. After everyone

finishes the problem, one question for the class would be, "What differences did you notice about when or how many people finished the problem?"

Potential Issues or Misconceptions when Solving the Magic Square

As stated above, solving the magic square purely by guessing is statistically unlikely, and it, of course, requires one to add numbers correctly. Other students in Education 130 were stuck using random number placements and required some hints to proceed. Notably, I believe the biggest roadblock would be not knowing the magic sum, which is 15 for the 3-by-3. Without knowing or discovering this sum, or potentially miscalculating it, a student will be stuck trying numbers and likely will not get anywhere meaningful. To address this potential concern, I may say, "What guesses that you have made have some potential use for this problem?" This pushes them to collect data on what does and does not work based on their guesses, and then use that to shape how they will later solve the problem. If they are confused by this, I might say, "Are there any numbers that work well with another number?" so that they recognize that some pairs of numbers are meant to be in the same line. Alternatively, if I saw some of their guesses previously, I could ask, "What problems did you have with this guess?", which, after they explain what went wrong, can be followed up with "How could you fix that problem?". These questions are not explicitly revealing about any particular method, but probe the student to explain their ideas and thinking aloud so that I can have them consider why something is the way it is.

Regarding situations where the student is unsure where to place a particular number, a hint regarding the "importance" of a number, whether in reference to its appearance in sums of the magic number, its value in comparison to both the mean and median of any general magic-square-appropriate sequence, or whether it is a variable in constructing such a general magic-square-appropriate sequence, among other reasons, would help to get students on the right track. For example, if students were placing 9 in the middle, a question for students would be, "What does placing 9 in the middle do to your diagonals?" This forces the student to consider the implications of a particular number's placement so they may try a different constraint (that may work better) and elicit their problem-solving capabilities. Of course, if the student is confused by what I might be implying, I would have them try solving the rest of the square, and then find all of the sums to show how the 9 inflates all the sums, while one or two rows or columns are much smaller. Once they do all the sums, I would say, "What do you think about these sums?"

Another concern is that some students are overwhelmed with keeping track of all eight sums. Of course, this will become a bigger issue for larger squares, and thus, helping these students keep track of everything will involve hinting at one of two strategies: placing 5 in the center, or writing out all triplets that sum to 15. An obvious hint is to try to constrain the problem: "What if we focused on one line at a time?" This helps put students back on track and think about the ways the numbers can be added together to their corresponding sums.

Another potential issue may arise if a student is dead set on placing a few numbers in a particular spot, such that while it may seem to work initially, it does not. A hint to the student here would be, "Are there particular combinations of numbers that don't work?" This hint helps encourage students not to focus on one idea and attempt other ideas, fostering some sense of creativity that may help or even be necessary to solve a problem, which is often a common necessity in higher-level math.

Practical Educational Furtherance of the Magic Square

Given the trove of solutions for a magic square and its high accessibility, questions naturally arise as to why a particular solution works while others do not. For an odd n -by- n magic square, students may wonder why the center is always the average of the numbers, i.e., 5 for a 3-by-3, 13 for a 5-by-5, etc. As they have found the magic sum by then, I would say, "Consider that number's relationship to the magic sum" or "Consider the number's relationship to the other numbers." If and when students complete the initial 3-by-3 magic square, several modifications can be made to elicit a new understanding of concepts. These modifications are as follows:

1. Create a higher-order n -by- n magic square.
2. Reason/Prove why the 3-by-3 magic square is unique up to symmetry and rotation.
3. Create or solve a 3-by-3 magic square that uses numbers outside 1 to 9.
4. Create a completely new method to solve it.
5. Try to tile a grid using a magic square.
6. Do Arithmetic and Linear Algebra Operations Work on Magic Squares?

Create a Higher-order n-by-n Magic Square

The purpose of this secondary assignment would be to entertain the question of whether their method is replicable over different magic squares, and, if so, whether it works for all n . This assignment is effectively a precursor to mathematical induction: one proves a base case of a problem, and conjectures that if the n th version of the problem is true, then the $n+1$ st version is as well. What they will most likely see is that unlike the 3-by-3 (one unique solution) and the 2-by-2 (no solution), the 4-by-4 and above have vastly more solutions, thanks to increased degrees of freedom in arranging the center.

Reason/Prove Why the 3-by-3 Magic Square is Unique Up to Symmetry and Rotation

Solving this question can be approached using several of the previous strategies: starting with constraints, reasoning with all eight sums of 15, and recognizing that each sum must have an odd number of odd integers. These constraints force the odd integers to be in the center and on the edges. The odd integers that are not five are forced into pairs, and those placements effectively require the even integers to be placed in only one configuration.

Create or Solve a 3-by-3 Magic Square that Uses Numbers Outside 1 to 9

The premise of this secondary assignment comes from Eperon (1959), where a 3-by-3 magic square can be created using just three variables: a , the number in the center; and x and y , which determine how far the other squares deviate from a .

$a - x$	$a + x - y$	$a + y$
$a + x + y$	a	$a - x - y$
$a - y$	$a - x + y$	$a + x$

Figure 4: Generalized Method to Create a 3-by-3 Magic Square

The base case of the 3-by-3 matrix is $a=5$, $x = \pm 1$, and $y = \pm 3$ (or $x = \pm 3$ and $y = \pm 1$). Normally, you want x and y to be positive, but the pluses and minuses help cover all 8 magic versions of the unique 3-by-3 magic square, either by flipping or rotating the square. We can create a magic square from an arithmetic sequence with difference d by setting one of x or y equal to d and the other equal to $3d$. The variable a is the median and mean of the sequence.

As alluded to earlier, 5, as the median and mean of the arithmetic sequence 1, 2, 3, ..., 9, is in the middle for this reason. This version of the problem forces students to consider why some sets of numbers can be arranged to make a perfect square, and why others cannot. Through this, they exercise pattern recognition, and by experimenting with several sequences of numbers, they can discover this general method for creating and solving a magic square using a valid numerical sequence. This method works for 4-by-4 magic squares, but it requires 6 variables instead of 3.

12	384	699
1052	365	-322
31	346	718

Figure 5: A magic square with $a = 365$, $x = 353$, $y = 334$, and a magic sum of 1095

Create a Completely New Method to Solve it

This assignment gets students to think outside the box, especially if they have already used several of the main strategies, ideas, and techniques. This helps promote advanced problem-solving and creativity, and whatever strategy they make up could be a unique approach to the 3-by-3 magic square that nobody else has ever thought of. Again, creativity and thinking outside the box are useful not only for higher-level math but also to approach problems in the real world.

Try to Tile a Grid Using the Magic Square

This is a different, but more visually based method of showing that the 3-by-3 magic square is effectively unique. Tiling a grid with the magic square introduces them to the fact that, although one can still have all columns and rows equal the magic sum, one fails to have the diagonals equal the magic sum without the 5 in the middle. These kinds of squares are called semi-magic squares. It also points further to the question of why 5 makes the magic square truly magic, which allows this problem to be followed up by one of the other problems listed here.

Do Arithmetic and Linear Algebra Operations Work on Magic Squares?

This section is the particular basis of my lesson plan and research. If one lets magic squares M_1 and M_2 have magic sums of s_1 and s_2 , respectively, and one uses an arbitrary constant c , $M_1 \pm M_2$ and $c \cdot M_1$ are both magic squares,

with magic sums of $s_1 \pm s_2$ and $c \cdot s_1$. This mathematically means you can create a linear transformation $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$, where $f(x) = c_1 M_1 + c_2 M_2 + \dots + c_n M_n$, for constants $c_1, \dots, c_n$ and magic squares $M_1, \dots, M_n$. Ward (1980) shows that the dimension of the resulting vector space is $n^2 - 2n$, where the basis of the magic square vector space consists of $n^2 - 2n - 1$ squares with magic sum 0, and a third basis: the square of only 1's. Matrix operations on magic squares are more challenging. While one can perform matrix multiplication between any two magic matrices, as Rose (1973) demonstrates using eigenvectors, the resulting matrix $M_1 M_2$ is, in most cases, only semi-magic because the diagonals do not equal the new magic sum. Rose also demonstrates that the inverse of a magic square, assuming it exists, usually is a semi-magic square, where M_1^{-1} has a magic sum of $1/s_1$. Some exceptions exist, including the inverse of any 3-by-3 magic square, assuming it exists.

More Applications Beyond the Problem

Given the numerous insights available before and after solving the magic square, and its various properties, the magic square serves as an exceptionally rich problem that leads into numerous mathematical topics suitable for elementary and middle school students, and even up to advanced high school students.

For advanced high school students, the magic square has applications to both discrete mathematics and linear algebra. In discrete mathematics, the method of writing out all triplets with sum 15 has connections to generating functions, which counts all potential sums given a set of restrictions; the formulas for the magic sum and total sum are both a result of mathematical induction; and the secondary problems allow students to think creatively, often a necessity to conjure a reasonable proof as to why a pattern or proof works. The mere concept of solving the magic square is itself a system of $2n + 2$ linear equations, and its ability to remain magic (or semi-magic) under arithmetic and scalar matrix operations allows entry into concepts such as using matrix operations to solve an augmented matrix, linear spaces, and finding eigenvalues and corresponding eigenvectors. Magic squares' uses in statistics and probability are also quite large. As stated before, the center of odd-sized magic squares consists of the average value of the given numbers, and a lesson could be created that involves the distribution of numbers larger or smaller than the mean/median being even, which, from a physics standpoint, allows the center of mass to remain in the middle. Calculating the probability for solving the magic square randomly – 0.0022% – is itself a very intricate probability problem, requiring the student to understand the sizes of both the sample space and event space. For number theory, another proof of the "ordinary"

and "break" vector methods, or even the regular proof involving the Latin squares, could lead to a lesson on modular arithmetic. The secondary tiling problem has uses in geometry, especially if it is part of a project.

Conclusion

The countless approaches to the 3-by-3 magic square, as well as larger versions of it, allow students and teachers alike to view this magic problem not only as a cool mathematical problem, but also as a tool of creative problem solving that harbors concepts ranging from Grade 1 addition to middle school algebra and geometry, and from high school statistics to college-level discrete math. Its equally rich historical relevance and theoretical inquiries provide extensive historical context for any enthusiastic teacher-researcher like myself who is willing to dive deep into the development of its approaches, and consider how students could, with basic mathematics and intuition, derive these approaches for themselves with enough time.

Lesson Plan

If the lesson plan is unavailable or not viewable below, please reference the following link to obtain access:
<https://docs.google.com/document/d/1K99w1m10YaJG1D4ymBO119Gq84dbPIQ75iGVI6xtrpI>

5E Lesson Plan Template

Cal Teach Student Name(s):	David Ni		
Mentor Teacher Name:	[REDACTED]	School/Room#:	Over Zoom
Grade Level and Subject:	12th Grade (advanced) (Note: its implementation for research purposes uses undergraduate students), Linear Algebra (Vector Spaces)		
Date & Time to Be Taught:	Saturday, August 1, 2026 at 9 AM and 2 PM CDT		

Lesson Source(s):	Eperson, D. B. (1959). Magic square patterns. <i>The Mathematical Gazette</i>, 43 (346), 273–275. https://doi.org/10.2307/3610655 Rose, D. M. (1973). Magic squares and matrices. <i>The Mathematical Gazette</i>, 57 (399), 36–39. https://doi.org/10.2307/3615167 Andrews, W. S. (1960). Magic squares and cubes (Repr. of the 2., rev. and enl. ed. from 1917). Dover Publ. SAWADA, D. (1974). Magic squares: extensions into mathematics. <i>The Arithmetic Teacher</i>, 21(3), 183–188. http://www.jstor.org.libproxy.berkeley.edu/stable/41188487 Barron, B. (2003). When Smart Groups Fail. <i>The Journal of the Learning Sciences</i>, 12(3), 307–359. http://www.jstor.org.libproxy.berkeley.edu/stable/1466921
Focus/Essential Question:	How can you make a vector space, and how can you determine and prove (or disprove) that a set of objects is a vector space?
Student Learning Objectives:	Students will be able to construct at least 2 real-valued generalized magic squares using the normal 1-9 magic square as a starting point through the use of matrix addition and subtraction Students will be able to identify at least 1 pattern that correctly results in a magic square Students will be able to define a vector space and its axioms in their own words via the Google Form exit ticket. Students will be able to identify a vector space by using the properties that determine whether a set is a vector space, which axiom(s) the set violates if it is not a vector space, and then demonstrate that understanding through the Google Form exit ticket problems.
Content Standards:	CCSSM N-VM-7: Multiply matrices by scalars to produce new matrices, e.g., as when all of the payoffs in a game are doubled. CCSSM N-VM-8: Add, subtract, and multiply matrices of appropriate dimensions MAA's Curriculum Guide to Majors in the Mathematical Sciences (CUPM): Recognize and use basic properties of subspaces and vector spaces
Student Prior Knowledge:	Students should be able to solve linear systems of equations. More importantly, students should, at the very least, be able to perform basic arithmetic.
Lesson Agenda for Your Students:	● Begin with solving a 3-by-3 magic square ● Try to make other magic squares using any numbers! ● Discuss students' methods in making those other magic squares with group members and the class ● Learn how these patterns connect to vector spaces!
Lesson Rationale:	By using the magic square as a matrix, and allowing students to manipulate it to generate new matrices, they will understand the axioms that define a vector space. By using the above content standards on matrix scalar multiplication and arithmetic operations on them, we can elevate their understanding to a more abstract manner. We ensure the achievement of the student learning objectives by having students identify certain properties of a magic square (Engage), understand how to construct other magic squares (Explore/Explain), identify a vector space, and understand what makes a vector space a vector space based on its defining axioms (Elaborate).

	The reason for using generalized magic squares is to satisfy the axioms of a vector space of generalized magic squares, since the set of normal 3-by-3 magic squares (using 1-9) cannot be a vector space, because it is not closed under addition or multiplication.
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Materials and Technology List:	Laptop (for Zoom and for local recordings in breakout rooms) Optional: Pencil, Paper For each group: One blank Google Slide and one blank Google Drawing At the end of the lesson: Exit Ticket via Google Forms All materials will be copied here and at the end of the Lesson Plan for ease of access. If links are inaccessible, please view the mock-up at the bottom of the lesson plan. Google Form: https://docs.google.com/forms/d/1zAO7GRY273yeuB_i0qQNGb3aGCGs_aE9deioepmNzd0/edit Google Slides: https://docs.google.com/presentation/d/1eaVVT1xW8Yr06hGCOspt4smT1_WN78W5hCV_68H-JsA/edit?slide=id.g3f5c0c89623_0_61#slide=id.g3f5c0c89623_0_61: Google Drawings: https://docs.google.com/drawings/d/1P83dMzKxwuBO_z_kjWjg3QjjqW3vA8yR6iGDbJJeJYw/edit
Preparation Tasks:	Prepare the Google Form for participants to fill out at the end of the lesson, which will include the problems in the elaborate section, reflection questions on students' individual and group work, and a place to upload local breakout room recordings (if necessary). Furthermore, create and prepare the Google Slides and Google Drawings for each group to use while working on the problem, and make sure links are available and accessible for each group Ensure that all audio/video recording features, aka Zoom Recording and OBS, are functional, and prepare a field notes document for your own research purposes. Boot up a Zoom meeting, ensure settings allow other participants to record, and ensure your computer properly saves both Zoom and OBS recordings.
Safety Concerns:	None

5E Lesson Plan, continued: Write the lesson plan as you would give it, providing yourself with a tool that summarizes flow and transitions, records age-appropriate wording for all instructions and prompts, and describes what you expect students to be doing and/or talking about at key points.

Lesson Title: Magic Squares as a Vector Space	Evaluate: Observe and adjust your lesson as you teach.
Engage: Activities that engage students' interest and build connections to their lives and prior knowledge.	Previous Experience and Baseline Learning
Assuming students followed directions, organized into their groups, and decided on who to record for their group, instruct them to begin recording. If students did not follow those directions, assign groups of 3-4 and randomly select one person from each group to record. Tell students that their names will be anonymized when editing the recording and that the videos are used for research purposes only. If students are unwilling to be recorded for research purposes, the teacher should note instances of student reasoning and group dynamics and reference the exit ticket responses to understand students' own thoughts. The group sizes are so that during the main activity, the problem is not so overwhelming, given the number of discoverable patterns and the ways they could fail. Thus, 3-4 people per group permits optimal collaboration while not leading to	Students are engaged if they are participating in whole-group or in their own group by actively helping each other find and use strategies to solve the magic square, such as generating new strategies or ideas to test or doing quick math to solve it, and anything in between. We expect prior knowledge to, at the absolute most, range from (at least for the group this lesson plan is used on, i.e. college undergraduates) partially sufficient knowledge on solving systems of 2 or more equations to understanding how to solve systems of equations using matrices. The

freeloading, whilst also ensuring students are less likely to become stuck. It is also more in line with and comparable to the group work study conducted by Barron in “When Smart Groups Fail”. The teacher should also begin recording on their end using both Zoom and OBS. Provide the Google Slides and Google Drawings to each group and instruct each group to assign themselves a number on Slide 3 (Individual #: First Name Last Name) As a quick engaging question, begin by asking the class, “Who is familiar with solving puzzles?” I will select 2-3 people and ask each of them, “Which puzzles?” or “What kind of puzzles?”, followed by, “What strategies do you use to solve them?” Write some of those strategies on the board as they say them. Begin by presenting the 3x3 magic square problem to students: “Well, today, we’re going to start with a puzzle, a math puzzle called the magic square! The magic square is an interesting math puzzle where you try to place the numbers 1-9 in a 3x3 grid, such that each row, column, and diagonal has the same sum. Your mission, for the next 10-ish minutes, is to try to solve it!” For 5 minutes, have students solve the puzzle on their own. After those 5 minutes, for the next 8 minutes, have students go into their groups, work together, and solve the problem and/or explain how they solved the problem. Instruct them to agree on one member to describe the strategies they used, what interesting aspects of the problem they found, and whether they solved the problem. Provide them with a shared Google Slide and/or allow them to use a Google Drawing to write down their ideas, work through the problem, etc. Furthermore, suggest that they each take up a role to ensure the productivity of the group (i.e., facilitator, recorder (either of their actual group and/or the group’s strategies, ideas, and failed attempts), presenter, math/square checker) After the 5 minutes of group discussion are up, temporarily bring students back together and have one member from each group discuss the above, and write/type everything they mention onto a shared screen they can see. Ask them to explain why they believe that strategy works (in any way, shape, or form) or why that pattern is there (in any way, shape, or form) Sentence starters should be given to students as below: “I/we solved the magic square by ____” “I/we recognized [pattern] when I/we ____ / because I/we ____.” “I/we tried [idea/strategy/pattern], but it did (not) work because ____.” Check that everyone is on board with how each strategy worked or did not work. Then, proceed by asking, “What if we replaced 1 with -4, 2 with -3, ..., and 9 with 4?” If they recognize that this is another magic square (whether by doing the math or realizing it is rotated), validate this discovery, and mention that there might not be so few magic squares as they might think. Tell students: “Now that we’ve discovered there might be more magic squares, I want you, as a group, to try finding more of these magic squares! You no longer have to use only the numbers 1-9! Use whatever math operations you like!” To support academic language development regardless of learning needs (including English language learners), Google Drawings helps students generate more visual notes of the magic square so that one can more easily see which rows, columns, or diagonals seem to have problems. Furthermore, if needed, the teacher should provide a quick	actual range is between solving systems of two or more equations and solving them using matrices. In a more reasonable/realistic application using the intended type of students (advanced 12th graders), the range of prior knowledge at absolute extremes would be between solving one-variable equations and solving systems of 2 or more equations. More likely, a majority would have prior knowledge of how to solve systems of 2 or more equations, with potentially one or two knowing how to solve a system of equations using matrices. However, given that the absolute minimum to solve the problem is basic arithmetic, revisiting prior knowledge may not be necessary, as long as students with more prerequisite understanding actively help those with fewer prerequisites to develop skills. If students are left behind, one should intervene to allow all students to participate equally. Patterns, strategies, or ideas that students may find when working on the normal magic square are as follows: Finding the magic sum either by realizing everything must have the same sum, or more mathematically using the formula. $n(n^2+1)/2$ for an n-by-n magic square Trying to put large numbers on the edges or corners. Recognizing that many pairs that add to 10 exist. Recognizing that 5 should be placed in the middle. Recognizing that an odd number of odd integers must be in each row/column/diagonal. Writing each triplet that adds to the magic sum. Using the Siamese (or related move/break) methods. Throughout the lesson, take note of the strategies that students used/mentioned in both the engaging question and when working on the normal magic square. Take note of how students reason about these strategies as well as the justifications they provide, whether from intuition, mathematical knowledge, etc.
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sketch of what “diagonals” (i.e., only the ones between opposite corners) are meant, and similarly for rows and columns. Hints can, and if needed, should also be used, including some listed below: “What do you see in your previous guesses that work and don’t work?” “How could you fix [particular problem]?” “What happens if you put [particular number] in [particular square]?” “What if you made this simpler?” “Are there particular combinations of number placements that don’t work?” If, after previous iterations of this lesson plan, time constraints persist, attempt to replace individual/group work on the normal magic square with a whole-group discussion of methods for solving it. Time: <u>Estimated 20-25 minutes</u>	
Explore: <i>Hands-on tasks designed to explore ideas and to develop skills together.</i>	<i>Focus, Involvement, Collaboration, Results, and Recording</i>
Remind students assigned to record to ensure that they are recording. If their recording breaks, they must notify the teacher IMMEDIATELY and have another group member record their group. Students will now go back to their initial breakout rooms with this new problem in hand. Instruct them to write down or record on the Google Slide every new magic square they create, and also note semi-magic squares, which are magic squares where at least one diagonal does not equal the magic sum. As they try to find patterns, enter each breakout room for around 4-5 minutes to ask about the patterns they found, the methods they used to find them, and ask questions that help elicit their thinking, observations, etc., using the prompts on the right. Simultaneously, observe each group’s Google Drawings and Google Slides to see what patterns they discover. The first three patterns are highly important for this lesson; the others are not as important, but are legal patterns to consider. Potential patterns that result in a magic square: Multiplying/dividing magic squares by any constant Adding/subtracting any constant to all entries in a magic square Adding/subtracting any magic square from another The following are also findable and valid patterns; however, they should only be taken note of for use in a later lesson if applicable: Multiplying magic squares by a matrix of all 1s (this results in a magic square whose entries are all the magic sum of the original magic square) Rotating/reflecting any magic square (From symmetrical properties of the magic square) Inverting any (invertible) 3x3 magic square (Comes from Rose’s paper) Constructing from an arithmetic series (More specific version of the general formula for a magic square) Constructing from a series arithmetically symmetrical around its mean/median (The general formula for a magic square) Raising the normal 3x3 magic square to an odd power (very complex, involves deconstruction into two separate matrices) Swapping symmetrical rows/columns (Effectively symmetry/reflection for 3x3; much different for higher-order magic squares) Potential patterns that result in a semi-magic square (note: while these are valid patterns that make a semi-magic square, they should only be taken note of for use in a late lesson if applicable): Multiplying a magic square with another one (not guaranteed to preserve the anti-diagonal) Raising a (non-normal) magic square to any power Raising the normal 3x3 magic square to an even power	Evidence of student learning will be counted if students can: Find several patterns that result in a magic square State some patterns that do not result in a magic square (or only a semi-magic square), Explain (in any way, shape, or form) why that pattern works to preserve magicness. Furthermore, students should be actively participating, whether by recording all patterns that worked, trying to find reasoning for why a particular pattern works, or finding new patterns, among other critical participation needs for learning. While checking in on groups, use the following questions/prompts: When first coming to check on a group, ask them to show you some magic squares they made and/or tested, and ask, “What did you notice when you were making these squares?” If they presented meaningful but unsuccessful attempts, ask, “How did you decide what to change so that the square is a magic square?” After their response, ask, “What happened when you tested that idea?” If there seems to be a quiet student, ask them, “What do you think your group should try next?” and then have the other group members only HELP the student as he executes that idea. If it fails, ask the quiet student, “Why do you think [idea] didn’t work?” If a group does not acknowledge a student’s suggestion, intervene and ask the student, “I heard you make a great point; could you maybe repeat it for the group?”

Swapping any rows/columns (This preserves the rows and columns, but not necessarily the diagonals)

Any arithmetic operations on semimagic squares (Semimagic squares are a superspace of generalized magic squares of the same size)

Once most groups appear to have found most of the patterns or we are approaching the time constraints of this section, tell students (or by Zoom broadcast), “We will be going back to whole class discussion in around 3 minutes. Please add your final patterns and/or explanations to your Google Slides and be ready to discuss your findings.”

During their group discussions, you can always suggest to some students to try copying a magic square (on Google Slides/Drawings) so that they are not caught up in the abstractness of making another magic square, instead allowing them to have a more visual idea of making one magic square into another (This is technically transformations, although this is not the talk of this lesson)

If limited on time (e.g., in a traditional class), have students save everything as is, and begin the rest of the lesson plan the next day. Otherwise, if working on a block schedule, take note of how much time remains for the Explain and Elaborate sections. If necessary, prepare to have students complete the exit ticket problems in the Elaborate section and their reflections separately as homework instead.

After 15-20 minutes, bring students back and continue as follows.

To support academic language development regardless of learning needs (including English language learners), hints can, and if needed, should also be used, including some listed below:

“Try using [insert student-mentioned strategy]!”

“Try just selecting numbers at first! See if they work!”

“What if you treated these squares as something simpler? What thing can you represent them as that is not so complex?”

Alternatively, the teacher should (at least attempt) to make a visual of the strategies or ideas to be more understandable for the rest of the group when a student suggests it and someone else is confused. For example, if someone mentioned adding magic squares together and someone is confused, instead of letting that student continue to be confused about it and follow along with the rest of the group, the teacher should step in and demonstrate that (Actually adding the two magic squares together and actually showing that the new magic sum is the sum of the two other magic sums). Alternatively, you can create a worksheet for students to fill out with these patterns; however, it would be hard to predict the order in which patterns are mentioned.

Time: Estimated 15-20 minutes

and explicitly ask the other group members, “What do you think of his/her remark?”

Students will be stuck if they do trial and error constantly without any improvement or progress towards finding an optimal setup. In this case, suggest they start with something simpler, use the normal magic square as a basis, and remind them to try arithmetic operations.

Furthermore, students may attempt to make semi-magic squares into magic squares, which rarely works. If such an event arises, validate their findings, but ask them to write it down and try to find other patterns instead of continuing to brute-force semi-magic squares

If students are confused, please refer to the hints on the left. If such hints are not relevant to the situation, then have students explain their thought process or point of confusion and proceed from there.

If students finish quickly (as in they find every single pattern and semi-pattern), have them perform their patterns on a 4x4 magic square to see if patterns still work. If students “finish” but did not quite find all of them, ask what patterns they found and suggest certain mathematical operations they could try to find other patterns.

If students find a pattern that was not thought of earlier, inquire further: “Oh, I didn’t think of that. Could you walk me through how you came up with that pattern?”, and ask them to present that pattern to the rest of the class later.

Teachers should also be ready for questions, ranging from clarifying a pattern (“Does the center square always contain the middle number?”) to having a completely new pattern you did not consider, all the way to difficulty or confusion (“Why does this pattern seem to work only sometimes?”)

For research purposes, in each breakout room, take explicit note of and discern between explanations and reasoning behind individual student ideas, group ideas, and group-influenced individual ideas so that clarity may be achieved in how groups help clarify or expand on ideas (or don’t). More specifically, ask the following to a group and/or a specific student in the group: “What patterns did you yourself

	find?” or “What strategies or ideas did you specifically try?”, followed by “What was the reasoning you gave as to back up this?”; “What patterns, strategies, or ideas did your group find or use?”, followed by “How did they explain why they tried this / why this pattern works”; “How did your group’s response to your patterns, strategies, or ideas help make your ideas clearer or more focused?” A similar set of questions is in the exit ticket; however, taking note of these ideas during thinking time allows the teacher to avoid students muddling these differences after a mind-intensive lesson.
Explain: <i>Students explain the phenomena they explored and discuss their different ideas and perspectives.</i>	<i>Participation, Reporting, Debating, and Evidence-Based Reasoning</i>
Before bringing everyone back, display the following rules of discussion for the class to see: “Think critically about how others think of a pattern.” “Be respectful about what others say.” “If you’re confused, ask! There’s no shame in not understanding something!” “Always try to explain your reasoning and thinking to others.” “Always try to keep the discussion going.” Additionally, provide the following sentence starters for students to use: “I/we saw [pattern] when I/we did/tried ____.” “What made you think of [pattern]?” (if students wish to ask about how others found this pattern) “I/we think [pattern] is correct/incorrect because ____.” “I/we thought of [pattern] in this way: ____” After bringing students back, instruct all students who were recording to stop recording and save the recording to an EASILY accessible place with an EASILY recognizable file name (Date_Name_Time_FirstName_LastName_Time) (e.g., August_1_2026_John_Smith_9AM.mp4). Have one member from each group mention any patterns they saw while creating new magic squares after you ask, “What did you notice about creating magic squares as you were finding these patterns?” Have them explain how they found this pattern, and how it makes a magic square. As they present the patterns they found, if they mention any of the above patterns that make a (semi-)magic square, inquire further: “What made you notice [insert the pattern they mentioned]?” For the non-highlighted methods, inquire with the following prompts: “Could you represent that [insert the method they mentioned] in a different way?” “How does this [insert the method they mentioned] connect to what someone else said?” “Who agrees or disagrees with this pattern, and why?” (have two or three students make their case before proceeding with validating or not validating the pattern) If student participation starts to slow down, ask, “What are some different approaches to think about a magic square?”, and ask, “How can we apply [insert mentioned approach]?” Once we approach the estimated time limit for this section or cover at least the first 3 patterns, the teacher should reiterate/summarize the patterns they discovered. If time is extremely limited, i.e., 20 minutes remaining at the start of this section, do a hard stop at 10 minutes, and if not all important patterns (the ones in the first set of patterns not highlighted in green) are covered, reference the contingency plan on the right.	Some misconceptions that may arise are that they mistake certain operations as ones that can create another magic square (when, instead, they made a semi-magic square or not one at all) However, one should NOT disregard these misconceptions, as it could hypothetically be used later in a class on subspaces (as normal magic squares are a subspace of magic squares, which are a subspace of semi-magic squares, which are a subspace of matrices) or, if time permits, in the use of polynomial vector spaces and determining whether an element is part of a vector space. When students share the ideas, make sure to take explicit note of the student’s reasoning, and whether it is their own or another group member’s. See how students improved their reasoning/explanation or built upon their group member’s reasoning/explanation. If specific students are sharing too much, which will be the case given that each group assigned a group speaker, the teacher should move to ask other students about the other patterns they found, or at least what ideas and strategies they tried, and why it worked or didn’t work. If some students seem to be avoiding talking, call on them and ask for their thoughts: “What do you think we can use these patterns to do?” or “Do you think there are other ways we could represent these patterns?” Given that students will not find every pattern, if time begins to run out and the core patterns that relate to vector space axioms are not all mentioned, ask: “Is there

To support academic language development regardless of learning needs (including English language learners), like before, the teacher should (at least attempt) to make a visual of the strategies or ideas to be more understandable to all in class. For example, if someone mentioned adding magic squares together, then try actually demonstrating that (Adding two magic squares and showing that the new magic sum is the sum of the initial two magic sums). Alternatively, you can create a worksheet for students to fill out with these patterns; however, it would be hard to predict the order in which patterns are mentioned. A word bank can also be provided: Pattern Magic Square/Sum Arithmetic operations (Addition, Subtraction, Multiplication, Division) Average Total To each one Time: <u>Estimated 10-15 minutes</u>	anything simpler that we can try?”, “Could [pattern] be broken down into smaller patterns?”
Elaborate: <i>Teacher-stimulated application and clarification of concepts, skills, attitudes, processes or terminology.</i>	<i>Demonstrated Understanding, Use of Skills, and Other Applications</i>
Formally introduce the concepts of a vector space as follows: Ask the following questions “You see that all of these (point to the actually successful patterns) patterns resulted in another magic square, right?” “Is a matrix full of 0s a magic square?” “For each magic square you made, did you find another magic square that could add to it to make 0?” Then formalize the definition of a vector space (in this case, it is a vector space of generalized magic squares): “A vector space is a nonempty set, say S, of vectors on which are defined two operations, called addition and multiplication by scalars (real numbers), subject to the ten axioms (or rules) I will list in a moment. These axioms must hold for all vectors $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ in S, and for all scalars c and d. 1. The sum of $\mathbf{u}$ and $\mathbf{v}$, denoted by $\mathbf{u} + \mathbf{v}$, is in S 2. The sum must be commutative, i.e., $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$ 3. The sum must be associative, i.e., $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$ 4. A zero object $\mathbf{0}$ is in S such that $\mathbf{u} + \mathbf{0} = \mathbf{u}$ 5. For each $\mathbf{u}$ in S, there is an object $-\mathbf{u}$ in S such that $\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$ 6. The scalar multiple of $\mathbf{u}$ by c, denoted by $c\mathbf{u}$, is in S 7. The distributive property works for scalars, i.e., $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$ 8. The distributive property works for vectors, i.e., $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$ 9. Multiplying scalars is associative, i.e., $c(d\mathbf{u}) = (cd)\mathbf{u}$ 10. A multiplicative identity exists, i.e., $1\mathbf{u} = \mathbf{u}$ Show that the set of magic squares is a vector space by mentioning and using student examples of each of the patterns they used above: 1. Using the pattern, “Adding/subtracting any magic square from another”, use a student example of that pattern 2. Take the student example used from above and reverse the order. 3. Take the student example used in #1 and introduce the normal 1-9 magic square 4. Using the above question “Is a matrix full of 0s a square?”, take a student-found generalized magic square and add the 0 matrix to it to show it is the same. 5. Using the above question “For each magic square you made, did you find another magic square that could add with it to make 0?”, use a student-provided answer to that question to show that.	If students are unsure about how to prove that something is a vector space, try asking the student about what he does or doesn’t understand about the box of things analogy: “So, you remember the box of things analogy I said right? What did I mention about what is in that box?” If they are specifically unsure about what to even do, but they understand the axioms, ask them, “What did I say about what you should use when proving these vector spaces?” If they don’t remember “variables”, say, “Well, you want to prove everything you make is also in the vector space, so what if you represented one thing in it as some sort of placeholder?” If they do remember “variables”, say, “Okay, think of them as a placeholder for whatever you want” Additionally, have them refer to the sentence starters and the word bank. Of course, if they do not tell me they are confused when I explicitly ask them, even if they are, then when they work on the given problems, I will ask them to talk me through their reasoning for a problem they already did or are currently working on: “Tell me how you knew this set was/was not a vector space.” For research purposes, collect their answers and reasoning behind the problems, and follow up with them if necessary.

6. Using the pattern “Multiplying/dividing magic squares by any constant”, use a student-found generalized magic square and the constant the student also used to show that.
7. Use the constant used in showing 6 and the two matrices used in showing 1 to show the distributive property
8. Use the constant used in showing 6, the matrix used in showing 6, and another constant a student suggests to show the distributive property
9. Use the same constants and square used in showing 8
10. Use any student-found generalized magic square and multiply it all by 1

“Now, what we are SUPPOSED to talk about is a little simpler than this, because we want to talk about vector spaces. It’s the same concept, but all objects are vectors. Everything we have shown here is a simple example that demonstrates that all generalized 3-by-3 magic squares are a vector space of their own. All of these axioms ensure that a vector space is **closed**. Basically, if you have a box of items and an operation, if one combines two items using an operation, then that new item must also be in that box. If it’s not, then it’s not closed, and it’s basically not a vector space.”

“However, when you actually prove vector spaces, we want to use VARIABLES. “ Proceed with using the constants x, y , and z , and filling two different magic squares (referred to as M_1 and M_2) with the variables a to i and the variables j to r , respectively, with respective magic sums of S_1 and S_2 , and show that $x[z(M_1 + y*M_2)]$ is still a generalized magic square, and manipulate it to show that axioms 1, 2, 7, 8, 9 are proven. Further manipulations can be made to show the rest.

Alternatively, you may follow more strictly the example shown by SAWADA, D. (1974). Magic squares: extensions into mathematics. *The Arithmetic Teacher*, 21(3), 183–188. <http://www.jstor.org.libproxy.berkeley.edu/stable/41188487>

“Now, does everyone feel confident about what makes something a vector space?”
(Address any issues or concerns as follows on the right)

Can someone explain to me why I had you all find these generalized magic squares and use those instead of just the normal magic square using 1 to 9?

Listen for student understanding of axioms that define a vector space. If they blatantly just state the axiom without backing it up, elicit that reasoning, “Can you show why (not)?”

Then, present the following problems in the exit ticket for them to work on, either in class or, if running out of time, as a post-lesson homework that will be discussed in the next lesson before moving on to new content. Students are expected to complete the Exit Ticket individually, and their responses will serve as the summative assessment:

Ask each student: “How would you define a vector space?”

For all of these problems, justify your reasoning by showing a counterexample that violates an axiom or justifying by using all 10 axioms (you can combine them for simplicity if you want)

1. Are all the real numbers a vector space?
2. Are all the positive real numbers a vector space?
3. Are all the integers a vector space?
4. Are all vectors $\langle x, y \rangle$ such that $x^2 + y^2 \leq 1$ a vector space?
5. Is the set $S = \{ \langle a, 2b, 3c, 4d \rangle \mid a, b, c, d \in \mathbb{R} \}$ a vector space over $\mathbb{R}$ under standard vector addition and scalar multiplication?

Optional problems:

6. Is the set of all single-variable functions $f: \mathbb{R} \rightarrow \mathbb{R}$ a vector space over $\mathbb{R}$ under pointwise addition and scalar multiplication?
7. Is the set of all indefinite integrals of continuous functions $f: \mathbb{R} \rightarrow \mathbb{R}$ with integration constant $C = 0$ a vector space over $\mathbb{R}$?
8. Is the set of all integrable functions $f: \mathbb{R} \rightarrow \mathbb{R}$ a vector space over $\mathbb{R}$?
9. Is the set of all differentiable functions $f: \mathbb{R} \rightarrow \mathbb{R}$ a vector space over $\mathbb{R}$?

10. A semi-magic square is a 3×3 matrix whose rows and columns share a common sum (diagonals are not required to match). Show that the set of all 3×3 real-valued semi-magic squares forms a vector space under matrix addition and scalar multiplication. If students were tasked with recording their group, instruct them to end the recording, go to the end of the exit ticket, and immediately upload the recording before they forget. If students were using pencil and paper for any activity, also instruct them to scan and upload their work into the same spot for uploading material. The teacher may stop recording here, i.e., at the end of the lesson With regards to supporting academic language development regardless of learning needs (including English language learners), please implement a word bank that includes the key terms that refer to the axioms that define the vector space: Word Bank: • Sum • Associativity (you can change the order) • Commutativity (you can swap two things) • Distributive (you can apply something to everything it applies to) • Additive Inverse • Zero • Scalar Multiplication • Closure Also, provide sentence starters in class and/or on the Exit Ticket: [Operation] validates axiom(s) [#(s)] because ____ This set is not a vector space because axiom [#] is violated when ____ Time: <u>Estimated 20-25 minutes</u>	
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Google Form: https://docs.google.com/forms/d/1zAO7GRY273yeuB_i0qONGb3aGCGs_aE9deioepmNzd0/edit

If the form is inaccessible, the problems inside are still the same as provided in the Elaborate section. The second section includes several research questions that the teacher may create at their own discretion.

Google Slides:

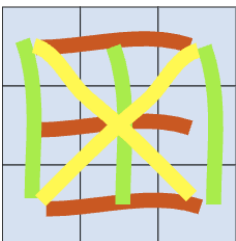
https://docs.google.com/presentation/d/1eaVVT1xW8Yr06hGCOspt4smT1_WN78W5hCV_68H-JsA/edit?slide=id.g3f5c0c89623_0_61#slide=id.g3f5c0c89623_0_61

If Slides are inaccessible, reference the following pseudo-mockup:

PART 1:

Traditionally, here are the rules

- Use the numbers 1 through 9
- Make a 3×3 grid
- Try to use each of the numbers only once
- Your goal: Have EACH row, column, and diagonal have the same sum



PLEASE NOTE: The following slides are split into GROUP and INDIVIDUAL slides. Please assign yourself an INDIVIDUAL NUMBER below:

Individual 1:

Individual 2:

Individual 3:
Individual 4:

INDIVIDUAL 1 SLIDE: Copy/paste all attempts you made at solving the magic square here
(Separate slides for each individual and a group slide where students may compile everything together)

INDIVIDUAL 1 SLIDE: Put any and all strategies, ideas, patterns, methods, etc., that you tried or used down here!
(Separate slides for each individual and a group slide where students may compile everything together)

GROUP SLIDE: When you find an answer to the magic square, copy/paste it below
(Only a group slide)

GROUP SLIDE: What are some similarities and differences between the completed magic square and your other attempts? (Only a group slide)

PART 2:
Are there other magic squares? Why Yes? Why No?

NOW the rules have changed
You NO LONGER have to use the numbers 1-9... You could use any number (even if it's irrationally complex). However, I suggest that you keep all numbers real. You also no longer have to avoid repetition of numbers.

INDIVIDUAL 1 SLIDE: Copy/paste all attempts you made at solving the magic square here
(Separate slides for each individual and a group slide where students may compile everything together)

INDIVIDUAL 1 SLIDE: Put any and all strategies, ideas, patterns, methods, etc., that you tried or used down here!
(Separate slides for each individual and a group slide where students may compile everything together)

GROUP SLIDE: When you find an answer to the magic square, copy/paste it below
(Only a group slide)

GROUP SLIDE: What are some similarities and differences between the completed magic square and your other attempts? (Only a group slide)

GROUP SLIDE: When you find ANY different magic squares, copy/paste it below
(Only a group slide)

Google Drawings: https://docs.google.com/drawings/d/1P83dMzKxwuBO_z_kjWjg3OjjqW3vA8yR6iGDbJJeJYw/edit
If Drawings is inaccessible, please view the following mock-up of the file.
In essence, if X represents each 3-by-3 square, the Google Drawings should look like this:

X X X X X Individual 1
X X X X X Individual 1
X X X X X Individual 2
X X X X X Individual 2
X X X X X Individual 3
X X X X X Individual 3
X X X X X Individual 4
X X X X X Individual 4

Methods

Participant Recruitment

Student participants were selected by posting targeted research advertisements on my Instagram story to 750 people, with 85% being alumni or current students at the high school I attended, and the other 15% being other friends or acquaintances I knew. I used the “Close Friends” feature on Instagram to broadcast the research advertisements, as well as direct messages via Instagram and Apple iMessages. I switched to and advertised a Google Form later on, which gave and requested the following information:

1. Details of the research that would be conducted.
2. Availability with a 2-week time frame.
3. Preferred mode of participation.
4. Comfort level and willingness to be recorded.
5. Request for anonymity and comfort level with their inclusion in a research paper.
6. Past coursework taken.

After around 3 weeks of continuous advertising, 14 people applied, 10 were selected, and 7 actually participated as described below.

Setting and Participants

Student participants in this study included 6 rising 2nd-year college students and 1 rising 1st-year college student. All participants are male and were comfortable with participating. The study was conducted between two 90-minute online synchronous Zoom sessions on August 1st, 2026, with 1 group of 5 participating in the morning session and the other two working together in the afternoon session. For future reference, I refer to the participants by the following pseudonyms as per participants’ requests:

Morning Session	Afternoon Session
Student A	Student J
Student B	Student K
Student C	
Student D	
Student E	

Descriptions of Participants

Students A through E and Student K are rising 2nd-year college students, and Student J is a rising 1st-year college student. All participants graduated from the same private high school as I. Students J and K both took Multivariable Calculus in 12th grade, and Student K took Linear Algebra as a college freshman. Students A through E took AP Calculus BC in 12th grade, but did not progress much further in mathematics in college, or it is unknown due to lack of foresight while developing the participation interest form. Students A through E also applied as a group and are a tightly-knit friend group. Students A through C, Student J, and Student K connected to Zoom via computer; Students D and E connected to Zoom via phone. Student E disconnected 30 minutes into the morning session before a majority of the study was underway, while Student D disconnected an hour into the morning session. All other students remained for the duration of their respective sessions.

Student A is majoring in a quantitative social sciences field. Student B is majoring in an applied business and operations field. Student C is majoring in a design and environmental studies field. Student D is majoring in a physical engineering field. Student E is majoring in a business and management field. Student J plans to pursue a major in the mathematical sciences field. Student K is majoring in an applied engineering field.

Confounding Variables

Because Students A through E are close friends, discussion was more likely to occur spontaneously in the morning group due to accustomed ease among the group, as opposed to a group of students who do not know each other as much, which was the case between Students J and K. As such, I noted these differences, and considered how these connections, as well as prior mathematical knowledge, would affect group success and collaboration.

I left Students D and E in this research as participants despite their not contributing much to the relevant research data, as their presence revealed several technological and social implications that will be addressed later. They did not affect the rest of the morning group's productivity except for exit ticket responses.

Group sizes are also of note, given that the group size ratio between the morning and afternoon group was 5:2 at the beginning, and 3:2 once the main portion of the study began. The afternoon pair was created to test a mini-experiment: could higher mathematical maturity suffice to solve the problem despite having one fewer student, and would group dynamics be noticeably different, either positively or negatively?

The biggest note is that all participants are male and graduated from the same all-boys private high school. As such, my overall generalizations about group work may not apply to most schools, but the overall supporting advice and suggestions are relevant to group success in the classroom.

Organization and Execution of Sessions

Participants in the morning session were a bigger-than-anticipated group of 5, while the two participants in the afternoon session worked together. The effective number of participants in the morning group, however, was between 3 and 4, as Student D was often distracted and Student E disconnected 30 minutes after the morning session started. For the relevant sections of the lesson transcript used, the morning session had 4 students; the afternoon session was the initial pair. I used a neutral teacher-researcher stance throughout both sessions.

Students' Goals

Participants were instructed to solve the original 3-by-3 magic square, first individually, and then as a group. This warm-up prepared students to consider the structure of a magic square by using intuition, preliminary patterns, and mathematical reasoning. Afterward, participants were shown another magic square that did not use the numbers 1 to 9, and with that in mind, they were instructed to find other magic squares, now with neither the restriction of the numbers 1 to 9 nor the restriction on numerical repetition. Furthermore, participants were instructed to work together and conjecture about the patterns behind these generalized magic squares. Their group discussions, reasoning, and revision of ideas, or lack thereof, that are used for solving the original magic square and conjecturing patterns that define generalized 3-by-3 magic squares are the main focus of the research.

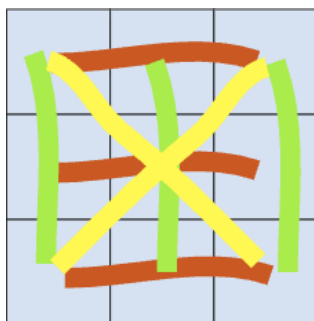


Figure 6: Provided Figure on Google Slides for Magic Square Rules

Provided Materials

I provided each group with their own copy of a Google Drawing, which consisted of several empty 3-by-3 squares, so that they can work on their own attempt(s) at a magic square while seeing how other students solve the problem. Additionally, Google Drawings has visual drawing tools that allowed multimodal, specifically visual, instructional use to help facilitate student reasoning. I also provided each group with their own copy of a Google Slides, which contained the general outline of the study and lesson, the parameters and rules of the magic square problem and generalized magic square problem, and ample space so that students may copy both successful and unsuccessful magic square-creating attempts and strategies, ideas, or theories from their Google Drawings onto the slides to act as a documentable source of student reasoning over time.

Google Drawings and Google Slides

The Google Drawings contained 40 3-by-3 squares made using Google Drawings' shape boxes. The space Google Drawings initially provided was insufficient, so I expanded it to 32-by-50. Each 3-by-3 is copyable, editable, and removable for students' convenience.

The Google Slides contained space for students to type out their thinking, strategies, ideas, etc., and space to paste their completed magic square(s). It lays out short but required rules about the magic square and generalized magic square, including a diagram, and asks conceptual questions, like "What are some differences between the completed magic square and your other attempts?" and "Are there other magic squares? Why or why not?"

Data Collection

Throughout both Zoom sessions, I recorded group conversations using both Zoom and OBS. I formatted the OBS recording to record the Zoom meeting, the digital Visual Studio Code TeX presentation, the lesson plan, and each group's Google Slides and Google Drawings in a combined, but separate split-view Google Tab. Had all participants shown up as instructed, I would have instructed students in the morning session to record their group's discussion using Zoom's recording feature to cover instances that neither my Zoom nor OBS recordings could cover. During both sessions, I typed field notes as commented-out text in the Visual Studio Code TeX file of student interactions with the problem and each other as they worked to find patterns that made magic squares. I noted how students found and explained their patterns to their group members, and how their group members used or ignored that information to progress through the problem and revise their ideas. At the end of each session, I included the final, group-edited copies of their Google Drawing and Google Slide, with minute-by-minute timestamps of all edits, as well as student responses to a Google Form containing the exit ticket, which asked students to reflect on both their individual thought process and how that thought process and individual understanding changed or expanded when discussing ideas with group members.

Exit Ticket and Uploaded Files

The Google Form containing the Exit Ticket contained a mix of rating questions such as "On a scale of 1 to 10, how often do you think your comments/ideas/strategies were considered or discussed by/with your group members?" and short response questions – such as "Are there any patterns that you thought could create a magic square, but didn't quite work? How did you adjust your approach after that?" – with example prompts: "Ex: I tried [this pattern], but I realized that _____. Instead, I decided to _____." The rating-scale questions were inspired by White et al. (2007) in the study on negative perceptions of group work compared to individual work. Participants could also upload screenshots or scans of any work not on their group's Google Slides or Google Drawings, as described below. Responses to the Exit Ticket are viewable in the appendix for future reference.

Google Slides and Google Drawings

Students did not use Google Slides as much as initially intended due to previously unforeseen technology issues and time constraints. As such, time stamps using Google Slides were not applicable for data collection. However, the Google Drawings were still a reliable reference to magic squares that students made, whether correct or not.

Field Notes

Field notes were taken in the same Visual Studio Code TeX file that was being presented to students via Zoom's present feature and the "Display PDF" PDF included in the Visual Studio Code LaTeX Workshop Extension. Field notes were left commented out so they would not be displayed to students and were used to double-check and cross-reference video recordings on the ideas shared, discussed, and revised, and other activities.

Data Analysis

I prioritized the OBS and Zoom video recordings from both sessions, spanning from the Engage to the Explore sections. Secondary data sources also included field notes, Google Forms exit ticket submissions, and all Google Slides and Google Drawings.

After each session, I immediately saved my OBS and Zoom recordings to a separate desktop folder for easy access. I also immediately reflected on all mishaps and considered improvements that could be made in future instances.

Video Editing and Transcribing

A day after the two sessions, on August 2, I downloaded Adobe Creative Cloud and Adobe Premiere Pro, uploaded both the Zoom and OBS recordings for the morning session in one editing project, and did the same for the respective afternoon recordings in a separate editing project. I spliced together the OBS recording – capturing the Zoom application itself; my digital, live-updating LaTeX PDF; and the group's respective Google Slides and Google Drawings as one split-view Google Chrome tab – with the Zoom recording. Zoom recorded audio more reliably than OBS. I then checked the file and included only the Engage, Explore, and Explain sections. I then selected the Captions and Graphics option, selected the entire spliced video and audio, and had Adobe Premiere Pro transcribe the recording. I exported the Premiere Pro transcript as a text file, and, since it only labeled speakers as "persons", I fed Google

Gemini 3.1 Pro Extended the text file and the synthesized recording file, split into two pieces to fit the 1-hour file time limit the model enforced. I then asked it to replace the “persons” tag with actual names for personal reference with the following prompt:

“While the text is mostly correct, the speakers are not necessarily correct or in the format I wish, so take the audio file for the first/second half of the recording, and edit the txt file. The people are [REDACTED], based on the Zoom call in the bottom right, best indicated by their Zoom border turning GREEN when they talk. Some of the names may not appear in this first half, but may appear in the second half. Please KEEP TRACK of WHO is WHAT VOICE. PLEASE MAKE the RESULTING TEXT EASILY COPYABLE.”

Given the limitations of AI, I still conducted a manual review afterwards of relevant sections I used for this research by going through the synthesized recording and correcting both the speaker and spoken words. I also assigned each participant an anonymous name during this review. While I did not note AI error rates in speaker identification, in the transcript, I estimated the error rate to be between 2% and 7%. If the transcribed text was inaccurate, I made my own preliminary transcription and repeated the respective section of the video until it was correct. I acknowledge that the final transcriptions may not be 100% accurate, but they are an accurate representation of student thinking and discussion.

Video and File Analysis

Using the (noticeably) inaccurate AI-reviewed video transcript, I watched the synthesized recording to make manual corrections to the transcript, assigned participant pseudonyms, found portions representative of student thinking and group collaboration, and found the relevant magic squares in each session’s Google Drawings file to incorporate into this paper. The groups’ Google Slides files could only be used to confirm who created which magic squares. Google Forms responses to the personal reflection questions were imported into Google Sheets, separated into Morning and Afternoon responses, and sorted alphabetically using participants’ pseudonyms. I replaced participants’ real names with their respective pseudonyms. The exit ticket responses had several participant-made typos, and obvious fixes were made in accordance with APA guidelines while preserving the participants’ original meanings.

When watching through the synthesized video, I included relevant, connected sections of the transcript into this paper for a context-based interaction analysis, and, on a second watch, conducted a thematic analysis while explicitly

noting and counting instances of co-construction, generalizations, students' reasoning, student interpretation, and both conjecture and idea creation or revision. Afterwards, I conducted a more thorough analysis of the transcript and video after reviewing relevant literature to make better connections between student reasoning, conjectures, revisions, and generalizations with concepts listed in Barron (2003), vector space axioms, passivity for the afternoon session, and the development towards generalizations as described in Ellis (2007).

The considerations for counting an instance of student thought for a parameter were as follows:

1. Co-construction: Do participants work together to generate a generalized magic square or pattern?
2. Generalizations: Does the participant make generalizations about the properties of magic squares using discovered examples that would support such generalization?
3. Student Reasoning: Do students use algebraic or general reasoning to find other generalized magic squares?
4. Student Interpretation: Does a participant use and interpret another participant's idea to bolster or create an idea?
5. Conjecture/Idea Creation/Revision: Do students generate conjectures and ideas, and do they use reasoning or others' ideas to answer or revise them?

Execution Issues and Comments

Recruitment Mishaps

Because I had publicly abandoned Instagram a week before I advertised this research project, my abrupt and unexpected return likely caused doubt regarding the advertisement's legitimacy. Additionally, I initially had interested participants send an email to a secondary email. My initial thinking for using a secondary email was to ensure that participation interest emails were not lost. I also underestimated people's general availability over the summer. These factors contributed to a suboptimal number of participants who expressed interest, with only 1% of people who saw the advertisement following through with emailing the secondary email. Even after revamping the advert with a much better Participation Interest Google Form and increasing the number of people who saw it to 750, the percentage of actual participants remained 1%. Additionally, some data on mathematical maturity had not been collected because the form was poorly created; it only asked for the form filler's completed mathematical coursework, but did not ask for that of potential group members.

Meeting Chaos

Despite clear, explicit instructions and reminders about the morning meeting's starting day and time (August 1, 2026, 9:00 AM CDT), as well as instructing them to make their groups of 3-4, I was forced to contend with only five of the eight initial morning participants arriving by even 9:15 AM; the other three did not show up. Furthermore, I underestimated potential technical issues when participants were trying to do the activity on their phones, as they were unable to access the provided materials during the lesson/study (Google Slides and Google Drawings). While this inadvertently made my job easier by grouping them all into one group, data collection was worse and not as streamlined as initially desired.

Additionally, given that these students were part of a friend group, instead of asking me for help or asking their group members about what they did, they decided to invite 2 other members of their friend group to "help". They were not of use to the group, however.

Technical Mishaps

I was experimenting with displaying instructions and student-provided strategies live in a TeX file, using the Visual Studio Code LaTeX Workshop Extension. I pre-drafted the questions and places within the file to put student-provided strategies, ideas, and patterns to be commented back into the file and the displayed, real-time-updating PDF. I also checked beforehand to see where errors occurred that prevented the PDF from updating. However, I failed to notice during the morning session that I made a different error, preventing the web view from updating. This rendered all of the sentence starters and posted student strategies completely unseen. Thankfully, since I focused mostly on individual and group reasoning and students did not need any academic language support, the sentence starters were not necessary. I figured out where the error was during the afternoon session, but the fact that it took several hours to find the simple error of not enclosing n^2 in $\$$'s was a major wake-up call.

Additionally, while reviewing my recordings, I noticed that OBS failed to pick up other participants' voices a third of the way into the recording; Zoom audio was used to replace it. However, this issue did not occur for the afternoon session, suggesting that I encountered an unknown OBS bug by chance.

0.0.1 Thoughts on Personal Performance

As I was watching through the synthesized recording, I found that I was surprisingly not as nervous as I thought I may have come across to the participants. I answered all questions as intended based on my lesson plan, and I seemed comfortably engaged while noting students' thinking and their progress. Despite the several issues and chaos that occurred, I appeared to handle them with surprising ease, pivoting to ensure everything within my actual control could function well enough.

Results

Similarities and Differences in Group Success and Productivity

Both groups were eventually successful in solving the normal magic square, generalized magic squares, and the patterns that generated magic squares, satisfying learning objectives 1 and 2. An immediate difference between the two groups was that the morning group, despite initially being larger, was talkative and was productively collaborating. The afternoon pair, however, was much less talkative and required a lot more prompting to cooperate and think together. Both groups got two of the simpler patterns: adding or subtracting a constant to each entry of a magic square, and multiplying or dividing a magic square by a constant. Both groups required a hint for the third-most important pattern: Adding magic squares together resulted in a new magic square.

The morning group demonstrated noticeable co-construction and algebraic thinking that developed into generalization of scalar arithmetic operations on magic squares. The afternoon group, however, saw less cooperation, with Student J doing much of the work and pattern discovery while Student K did not consider most, if not all, of Student J's ideas.

Morning Group Success

Initial Magic Square Problem

Students initially had trouble working on the normal 3-by-3 magic square individually, as many tried using guess and check, making squares like

$$\begin{bmatrix} 9 & 8 & 1 \\ 2 & 6 & 5 \\ 7 & 4 & 3 \end{bmatrix} \text{ (Student C) and } \begin{bmatrix} 9 & 6 & 5 \\ 1 & 3 & 8 \\ 2 & 7 & 4 \end{bmatrix} \text{ (Student A).}$$

However, they quickly began sharing ideas once instructed to work together:

Student C: Does this have something to do with the number 45?

Student B: No, it's the sum; they should all add up to 15.

Student A: They sum up to 15? I've been trying to figure that out.

Student B: The sum? Well, I just added up all the numbers and divided by 3.

Student D: Does it have to be 15, or could it be another magic number?

Me: Well, if you think about it, maybe you'll understand. You do understand what it means for each row to have the same sum, right, let alone the columns and diagonals?

Student B: Oh, they all add up to the same number.

5 minutes later, Student B managed to solve it, and explained his reasoning: "They all had to add up to 15, because if you added them all together and divided that by three, it equaled 15. You needed the sum of three numbers to be 15, and then five had to be in the middle. He gave us that clue because five is in the middle; all the numbers one through nine." (For context, Student B previously gave an incorrect answer, and I made the remark that "I know if it's wrong just from looking at one number.")

The ideas that students exchanged had a noticeable effect on how students approached the normal magic square. To the exit ticket question, "Describe a moment during breakout rooms when a group member explained their strategy or pattern discovery. What was their idea, and how did it differ from your initial way of thinking about the

problem?”, Student C noted how “Student B mentioned his way of thinking, and it differed from mine because he opened my eyes to how to understand and break down the way the sum of 15 is the magic number.” Initially, Student C “approached it by guess and check, but afterwards, my opinion changed by using a better stepped approach.”

Patterns of Magic Squares

At this point, Student E left for personal reasons, and Student D was visibly distracted, bringing the total number of active participants for the morning to 3. After showing students that subtracting 5 from every entry created another magic square, I instructed them to find even more magic squares and the patterns that define them (i.e., transformations or their corresponding vector space axioms, neither of which I said explicitly, only implied). Immediately, students begin suggesting ideas and testing them.

Student A: Student C, do you have any thoughts about how to start this?

Student C: Multiply everything by 3?

Student A: Okay, of the correct solution?

Student C: Yeah, let’s see what happens. [silence] Everything has to be equal to 45 (the magic sum).

Student A: I feel like if we have the correct solution, can’t we just basically add as long as we add the same number to all of them? Does multiplication work, Student C, if you multiply everything by three?

Student C: I’m not sure, I’m just testing it out.

Student A: Gotcha. I think you can add or take away five (the example I showed them) or add a hundred to every single thing, and it should still be the same as long as you’re adding the same number to all of them.

Students A and C are productively cooperating, coordinating to figure out whether their ideas and actively taking up and testing ideas. Student C’s “multiply everything by 3” idea and Student A’s “add a hundred to every single thing” idea work and create two examples of a successful matrix transformation, with Student C’s representing scalar matrix multiplication, and Student A’s representing addition. These two are good examples of two particular vector space axioms, namely that scalar multiplication and, at least partially, vector addition within a vector space result in another vector in the space. However, I emphasize the partial nature of addition, as they do not explicitly consider “add

a hundred to every single thing” as a magic square with only 100 as its entries.

Notice as well that the students are already beginning to conjecture about generalization (“Can’t we just basically add as long as we add the same number to all of them?”), which is an interesting early demonstration of algebraic justification leading to better generalization (Ellis, 2007). This occurrence resulted from what Barron (2003) notes is the importance of joint attention; both the speaker (Student C) and listener (Student A) contributed to generating this initial generalization. Over time, they will make this generalization more concrete.

Student A: Okay, Student C, I think I figured one out. You can add 100 to everything, so instead of it being one through nine, it’s just 101 through 109.

$$\begin{bmatrix} 108 & 101 & 106 \\ 103 & 105 & 107 \\ 104 & 109 & 102 \end{bmatrix}$$

Student B: Student A, what do they [the sum of the numbers] add up to?

Student A: 1045. No, 945.

Student B: So, for the magic square, when you have 102, 105, and 108, it would add to 315. So, it’s the same core principle. If we put 105 in the middle, what you do is find which one is the mean and you put that in [the middle].

Student B used what he understood about magic squares in the initial magic square, specifically the magic sum and middle number, to make another generalization: “Find which one is the mean and you put that in [the middle].” Student B’s interpretation of Student A’s discovery helped enable his own generalization, using co-construction in the process. Notice how Student B unknowingly mapped a magic square to the value of the middle entry and inadvertently used function language without knowing.

Student A: Sounds great to me.

Student C: (to me) Are we correct?

Me: Well, what do you think?

Student B: Well, if you take the mean or the number in the middle of the data set of the nine numbers that are all

within one of each other, and then you try to add up to whatever the sum of all the numbers is divided by three, then that would always work as a magic square. [pause] So is the key like having a data set of nine numbers that are all one away from each other?

Here, Student B revises and formalizes his generalization to note that the total sum also follows a generalized pattern. In the process, he takes up another conjecture about the spacing of numbers. What he did not realize yet is that the others' generated magic squares already answered his conjecture.

Me: That's one way to think about it.

Student B: What if it's [the separation between numbers] not just one?

Student C: (to me) I figured another one out. You just multiply them all by three, and they all have to equal to 45, and 15 is the middle number.

$$\begin{bmatrix} 24 & 3 & 18 \\ 9 & 15 & 21 \\ 12 & 27 & 6 \end{bmatrix}$$

Student B: So it can be more than just one apart for it [a sequence of numbers]. [pause] (to me) Is it just that if you have a set of nine numbers, you put the middle number of that set in the middle of the magic square, and then you try to make every sum equal to the sum of the nine numbers divided by three?

Me: That's one way to think about it.

Notice how Student C's comment about multiplying the normal magic square by 3 prompted Student B to realize that, to answer his own conjecture, the numbers did not have to be separated by one; it could be separated by 3, using Student C's idea. He simply required another discovered magic square to discover the answer. Student C's reasoning and Student B's conjecture, although now proven incorrect, combined is another example of co-construction, which now generates his "not just one" spacing conjecture. This was a very powerful instance of student reasoning being altered by group discussion. Student C made another example of scalar matrix multiplication and thus the scalar multiplication axiom of the 3-by-3 magic square vector space, which will help translate their ideas into a future generalization.

Student B: I think it can work with fractions as well.

Student A: If you multiply all the numbers by 4, it adds up as well.

$$\begin{bmatrix} 32 & 4 & 24 \\ 12 & 20 & 28 \\ 16 & 36 & 8 \end{bmatrix}$$

Student B: Yeah, it works with fractions. I divided all the numbers by nine; it all added up to one and two-thirds okay, which is what I think happens if you divide 15 by nine, so that's also pretty interesting.

$$\begin{bmatrix} \frac{8}{9} & \frac{1}{9} & \frac{2}{3} \\ \frac{1}{3} & \frac{5}{9} & \frac{7}{9} \\ \frac{4}{9} & 1 & \frac{2}{9} \end{bmatrix}$$

Student B: (to me) Do you think there's like a fundamental sequence that every other sequence is based off of?

Student B's question again points towards how algebraic thinking leads to better generalization. Armed with his interpretation of data and findings of Students A and C as well, combined with his own formulated and revised conjectures and respective answers, he combined them all while referencing the numbers 1 to 9 as a base "sequence" – the other students had performed operations on the basic magic square, which better explains why he may have considered 1 to 9 as a potential base sequence – to make this conjecture. Although his question started to deviate from the three important patterns I used to introduce vector spaces, I understood his idea still had potential if the lesson plan had a different focus. Also of note was Student B showing that scalar division works as well, which further reinforced the scalar multiplication axiom for vector spaces.

Me: Possibly. You know Students A and C are saying that multiplying the square by 3 or 4 works, and Student B is saying that it works by dividing everything by nine. Do you think there's any connections between those?

Student B: I think, as long as the sum of the nine numbers divided by three is equal to the sum of the rows, and you

have the middle number of the nine numbers in the middle of the square, then it will work as a magic square. Wait, I wonder if the numbers are uniformly apart from each other. Actually, that's something to test. I'm going to go test it right now. What if the numbers aren't uniformly apart? We've been testing like 1 to 9 or 2, 4, ..., 18, but what if we try 1, 2, 4, 5, 7, 8, 10, 11, 13 randomly, and they're not uniform. Would it still be able to work if we add them all together, divide by three, and we put the median in the middle?

Student B's next conjecture that numbers did not have to be uniformly apart points to the generalized method to create a 3-by-3 magic square (as seen in Figure 4), and, had students known such a generalized method existed, it would have confirmed Student B's conjecture, albeit confined by the given restrictions on the square itself.

Me: It sounds like a good idea. Why don't you try it out? (I repeat the same statement as my previous statement).

Student C: Is there a chance that if it's one through nine in the box that you can multiply, divide, add, or subtract by any number to each one, and it'll always be a magic square?

Me: Possibly, but can you show that that's the case?

Student C: Student A, so we already have multiply by three and four, should we add by a number?

Student A: We added by 100, and that worked, so I think you can look at the five example [the example I showed them to introduce general magic squares]. You can add and multiply. And as long as you add or multiply both by the same number, it should work.

Student C: And you can divide because we did the fractions.

Student A: Yeah. You can divide as well.

Student C: So we need subtraction.

Student A: Yeah, subtraction would work too, so would addition, multiplication, and division. As long as you have the basic solution that we solved for one through nine, and as long as we do the same thing to every number, I think it works.

Student B: Once you have the basic series, you can modify it in any way, and it will remain a series.

Notice that Students A and C have begun to generalize the scalar arithmetic operations as legitimate patterns

or transformations that generate another magic square. These generalizations formally connect to both the vector addition and scalar multiplication axioms, and helped revise their initial generalizations. After I reiterated what magic squares and operations they found that worked, they referenced and interpreted their initial discoveries to make their generalization, which now not only included addition, but also multiplication, division, and subtraction. At this point, I had students share and summarize what patterns they found to pinpoint what else we needed to discuss.

Student C: Are you able to multiply every number by three and then add or subtract? Can you do two operations on one [square]?

Student C's question is similar to a linear algebra concept, notably linear transformations. I found the idea relevant, but unfortunately, time was too limited to consider that idea in the moment. It is, however, worth noting that they were effectively discussing the requirements of a linear transformation, i.e., additivity and homogeneity. Instead, I used it to steer them towards considering adding magic squares:

Me: What if he represented that five as something else?

Student C: Like n ?

Student B: Yeah, like n or x .

While that was not my intended direction, they interpreted it differently and considered not adding just a single constant, but adding a variable. Notably, their use of variables is another layer of formalization of scalar operations on magic squares, and thus another connection to the vector addition axiom.

Student C: The series from one plus n or subtracted by n .

Student B: So it could be like if you have a series of one through nine and it's one plus n through nine plus n , then it would be the sum of every row and column would be $15 + 3n$, [pause] so that could work for any number.

Me: I also want you to consider, what if we use 1 or 2 of the same number in a magic square?

I intended to use the concept of adding a constant to each number, and have them consider it as a magic square instead. It immediately prompted them to consider a matrix of 1's, which is a valid magic square. Student B also remarked that "it would still be uniform distance; the uniform distance would just be zero." Regardless, it took several tries to get them to consider adding two magic squares more literally. However, I find the resulting conversation still important to consider.

Me: What if that constant wasn't just a constant?

Student B: It could be any number like you just said, Student A. That would still work if it was like [unknown].0.

Student A: Yeah, it could be π , it could be an imaginary number.

Me: Do you notice how adding the same constant to a magic square is the same as adding this matrix full of the constants to that magic square is the same thing, right? And the matrix with that constant is also a magic square, right? What if you try to add these magic squares together?

Student B: So it would be adding every magic square n through the magic square n plus infinity. And then you would add everything up until n plus infinity, so it would be a magic square of n plus infinity.

Me: What you're saying is technically correct, but it's still technically a constant. I want you to try taking the original magic square, rotating it by 90 degrees, and then adding that rotated version to the original.

Student B: [After adding them] So, they would all add up to 30.

$$\begin{bmatrix} 8 & 1 & 6 \\ 3 & 5 & 7 \\ 4 & 9 & 2 \end{bmatrix} + \begin{bmatrix} 4 & 3 & 8 \\ 9 & 5 & 1 \\ 2 & 7 & 6 \end{bmatrix} = \begin{bmatrix} 12 & 4 & 14 \\ 12 & 10 & 8 \\ 6 & 16 & 8 \end{bmatrix}$$

Student A: Are we combining squares?

Student B: Yes, and it works, but you get multiples of the same number. You get two 8's, two 12's, so it's interesting, because that's not all uniformly separated.

Student C: So what's the pattern here?

Student B: Yeah, what is the pattern here? [pause] Is it because eight repeats twice and then on the opposite side, 12 twice?

Me: It could lead to one way to make a magic square, but I want you to consider the implications of what you got being a magic square.

Student B: Are you saying that any sum of two magic squares would also be a magic square?

Me: Possibly. Maybe you could test it out.

Student B: Perhaps, if you keep rotating it, or we could take that one, that something that we got, and we could add that to the original magic square. [wait] Yeah, it works as well; they all add up to 45, and now we have three repeats.

$$\begin{bmatrix} 12 & 4 & 14 \\ 12 & 10 & 8 \\ 6 & 16 & 8 \end{bmatrix} + \begin{bmatrix} 8 & 1 & 6 \\ 3 & 5 & 7 \\ 4 & 9 & 2 \end{bmatrix} = \begin{bmatrix} 20 & 5 & 20 \\ 15 & 15 & 15 \\ 10 & 25 & 10 \end{bmatrix}$$

While I succeeded in having them realize adding magic squares works perfectly fine, Student B realized that the numbers used in a sequence did not have to be distinct; they could be repeated. This challenged and falsified his initial conjecture that if you used a sequence to make a magic square, the sequence had to be uniformly separated.

Afternoon Duo Silence

Initial Magic Square Problem

The pair for the afternoon had as much trouble solving the normal magic square, which was against my initial hypothesis. Student J found the magic sum mathematically, using the sum formula $\frac{n(n+1)}{2}$ to calculate a total sum of 45 and divided it by 3 to find the magic sum of 15. Student K, however, only did trial and error. Working together, they seemed to be off track: both initially thought the magic sum was 16. They also tried placing the larger numbers a knight's move away, drawing from their familiarity with chess, as well as their novel "donut" method, which involved pretending the center square did not exist at all. Although they solved it 10 minutes faster than the morning group, they found the solution mostly by luck, as Student J noted in his exit ticket response; they reverted to putting 5 in the middle, and they kept using trial and error to solve it.

Patterns of Magic Squares

While this pair is not the main focus of the research, I think it is important to note that their cooperation in this section was noticeably lower. I had to constantly break the minutes-long silence to ask them to talk and work together to find generalized magic squares. As such, less transcript data was available; in fact, the transcript length for the afternoon pair was only 11 pages, while that for the morning group was 26 pages. Both were formatted the same way. Instead, I partially narrate the pair's work, noting the silence between bits of discussion.

Regardless, the pair did not collaborate for the first 3 minutes before I gave my first reminder. Another minute passed without any conversation before I deemed it necessary to probe students' reasoning:

Me: What made you come up with that [matrix]?

$$\begin{bmatrix} 2a & 7a & 6a \\ 9a & 5a & 1a \\ 4a & 3a & 8a \end{bmatrix}$$

Student J: You were talking about translations, so this is just another translation.

It became clear that Student J interpreted my example of subtracting 5 from the original magic square as a translation. The terminology is awkward in a way, as you cannot “translate” a matrix in a physical sense, but the concept certainly does work: you “translated” each entry of the magic square down by 5. In a similar sense, he decided to apply it to multiplication, but, unlike the morning group, he immediately went for a generalized concept by using a variable as the scalar constant, thus connecting to the scalar multiplier axiom. I was surprised that Student K did not pick up on this idea as quickly as Student J, partially because, given he took linear algebra in his first year of college, I thought he might pick up on a pattern quickly.

Unfortunately, the silence continued for another 3 and a half minutes, after which I asked Student K what he was trying; he was still using guess and check. I told him that would be tedious, and “maybe try like using the original Magic square as a reference, sort of like what Student J was doing with the one up top.” Meanwhile, Student J took notice of my attempted redirection, or perhaps on his own, updated his generalized matrix to allow addition with the

variable b , thus incorporating the addition transformation and its related vector addition axiom:

$$\begin{bmatrix} 2a + b & 7a + b & 6a + b \\ 9a + b & 5a + b & a + b \\ 4a + b & 3a + b & 8a + b \end{bmatrix}.$$

2 more minutes of silence ensued before I intervened again, reminding them to try working together. Finally, 15 minutes later, 10 minutes of which were silent, they were working together. At this point, no reasonable instance of co-construction even occurred; Student K continued with his guess-and-check method and did not take up anything Student J explained to smooth out his process. Student J explained the above generalized matrix to Student K, but noticed that Student K generated a magic square that did not follow that format:

$$\begin{bmatrix} 14 & 33 & 3 \\ 17 & 11 & 22 \\ 19 & 6 & 25 \end{bmatrix}.$$

Student K: The way I started it is I decided it's going to add up to 50, and then I made the diagonal. So I did 25, 11, 14. And then I kind of figured it out from there.

His strategy was still mostly trial and error; he did not necessarily have a rigid pattern to describe it, nor was it consistent; the other magic square that he made by then was not a magic square at all:

$$\begin{bmatrix} 6 & 15 & -1 \\ 5 & 4 & 11 \\ 9 & 1 & 10 \end{bmatrix}. \text{ The anti-diagonal } 9, 4, -1 \text{ does not sum to } 20.$$

2 more minutes of silence ensued before I asked who was setting up equivalence equations:

$$c + f = g + h, d + g = b + c, a + b = f + i, a + d = h + i, f - b = h - d$$

Student K didn't seem to get anywhere with this, however, despite my hopes and anticipations that he might solve for

them in a matrix, and thus find a valid parameterized solution and basis. Student J, however, appeared to be inspired by this and wrote down all the sums to be accounted for:

$$S = a + b + c = d + e + f = g + h + i = a + d + g = b + e + h = c + f + i = a + e + i = c + e + g.$$

He based it on this magic square full of variables:

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$$

He then put them all into another square for easier viewing:

$$\begin{bmatrix} S & a + e + i & c + e + g \\ a + b + c & d + e + f & g + h + i \\ a + d + g & b + e + h & c + f + i \end{bmatrix}$$

Meanwhile, another 2.5 minutes of silence ensued before Student K realized they were completely stuck. What is surprising is that neither one of them seemed to consider, at least explicitly, that the square full of sums was itself a magic square, effectively coming from multiplying a magic square by a matrix of 1's:

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

Instead, they saw it as a placeholder to consider how changes in variables affect the other sums. At the time, I thought they knew that that was a magic square, and did not press further. However, after re-watching the video, I realized that was not necessarily the case. In this moment, I hoped that Student K would use prior knowledge to consider solving the systems of equations with these matrices. He did not; "I'm not sure if that helps to get like a whole pattern." At this point, we were running out of the allotted time, and the best I could do was ask Student K to try adding his magic

square to the original magic square to expose them to the more explicit generalization of adding magic squares, and then move on to introducing vector spaces.

Over the course of 27 minutes of “group work” and “collaboration”, they were silent for half of that, mostly working separately. While Student J actively explained his ideas and reasoning to Student K, Student K mostly nodded and rarely attempted to integrate that into creating his magic squares. Of course, it is hard to analyze the instances of algebraic thinking that translated to generalization: Student J took my initial example and immediately generalized it; Student K never did any noticeable algebraic thinking. Regardless, no meaningful co-construction or active, sustained collaboration arose; Student J had done noticeably more than Student K.

Total Instances of Group Work Dynamics and Progress

The table below counts instances of group work dynamics, such as algebraic reasoning, conjecture writing and revising, interpreting others’ ideas, co-construction, and generalization.

Types of Group-work Dynamics	Morning Session	Afternoon Session
Co-construction	3	0
Algebraic Reasoning (or General Reasoning)	4	2
Generalization (Or Attempts at Generalization)	5	3
Instances of Interpretation	5	2
Instances of Revising/Creating Conjectures/Ideas	7	1

Table 1: Instances of Each Parameter

Notice how the participants in the morning section engaged in many more instances of all the above parameters. While this may certainly be attributed to various factors, including different group sizes, potential stranger mentality among the afternoon pair compared to pre-disposed comfort between the morning group members, or personality differences, the morning group succeeded more in creating conjectures and turning algebraic reasoning into generalizations. Meanwhile, nearly all instances of the parameters in the afternoon session came from Student J; Student K had only one instance of generalization when I had him add magic squares together.

Comparison of Student Responses Rating Scale Responses

The table below represents the average responses to the four personal reflection rating questions in the exit ticket. 10 represents “Very ___”, while 1 represents “Not at All___”:

Questions (Scale of 1-10)	Morning Average (N=4)	Afternoon Average (N=2)
How successful do you think your group was in finding other magic squares?	7.25	7.5
How often do you think your comments/ideas/strategies were considered or discussed by/with your group members?	7.75	6.5
How often did you feel that someone was not contributing to your group?	6.25	3
How much effort would you say you put into the group activities compared to the rest of the group?	6.5	4

Table 2: Average Responses to 1-10 Rating Questions

Statistically, these results do not reflect the true situation. The morning group’s responses to the third and fourth questions are inflated due to Student D’s lack of participation. The sample size was also small, and scores were subject to participants’ personal “interpretation” of the scale. It also does not cover the range of responses, which are covered below:

Questions (Scale of 1-10)	Morning Range (N=4)	Afternoon Range (N=2)
How successful do you think your group was in finding other magic squares?	7-8	7-8
How often do you think your comments/ideas/strategies were considered or discussed by/with your group members?	7-10	4-9
How often did you feel that someone was not contributing to your group?	5-8	2-4
How much effort would you say you put into the group activities compared to the rest of the group?	5-9	2-6

Table 3: Range of Responses to 1-10 Rating Questions

What should stand out is that the afternoon duo had noticeably different impressions regarding the consideration of each other's ideas and the effort they put in relative to others. Student J gave the responses of 4 to question 2 and 6 to question 4, while Student K gave the responses of 9 to question 2 and 2 to question 4. Their varying responses demonstrated a disconnect in their group collaboration due to a lack of co-construction and of explaining and sharing ideas; instead, they mostly sat in silence, and Student K mostly nodded in approval of Student J's generalizations without thinking further about it. Meanwhile, the morning group appears to be more in agreement in their responses; extremes in the data were Student B's response of 10 to question 2, and Student A's response of 5 to question 4.

Discussion

I believe my research has several parallels to that conducted by Barron (2003); the morning group exercised a lot of co-construction of solutions and "establishing joint attention". Meanwhile, the afternoon pair was, although not necessarily engaging in rampant "skipconnecting" – the ignoring of other students' ideas and focusing on their own –, less intent on spontaneous discussion and co-construction; rather, they were what Mulryan (1992) describes as "on-task at work but not engrossed", or "working on approved tasks" but not as "engrossed", as in they are not as

“actively discussing or interacting with other students on an approved task”.

Factors Contributing to Morning Group Success

The biggest factor contributing to their success was the spontaneity in productive collaboration. Even though two of the participants were unable to participate thoroughly due to technical limitations, and one of them was “off-task” – specifically “noninvolved and disrupting the work of other students” (Mulryan, 1992)– the other three were “on-task engrossed”, actively working on the problems at hand while cooperating immensely. Exit ticket responses noted how one student’s way of thinking “differed from mine because he opened my eyes to how to understand and break down the way the sum of 15 is the magic number.” Student B noted how he “was initially visualizing/approaching the magic square as a linear set of numbers. However, after hearing my peers explain their method, my understanding changed because I realized that the set of numbers did not need to be uniform and expanding in an increasing order but could take on many forms as long as they met the criteria of the ten [vector space] axioms.”

These students’ exit ticket self-reflections were proof of the effective and intertwined “co-construction of understanding” taking place: in the initial magic square problem, Student C’s question prompted Student B to answer with a useful answer about the sum, which, in turn, helped the other students reconsider and refocus their attention on the magic square with the magic sum of 15 in mind. Students A and C’s discoveries that multiplying entries of a magic square by 3 or 4 resulted in a magic square led to Student B making a useful note from their discoveries that of the numbers used in the magic square, the middle one must always be in the middle, and that you could also divide the magic square entries by a constant to get another magic square. These discoveries led Student B to consider the possibility that “the numbers aren’t uniformly apart”, leading to lots of testing. Similarly, Students C and A worked together and validated each other’s patterns to come to the generalization that scalar arithmetic operations on matrices work: “I was initially confused with Student C’s idea about multiplying the squares by each other, but it ended up working out. He showed me an example to explain it to me.”

In the two instances where I did not succeed in steering their thoughts to adding magic squares, they inadvertently made progress towards actual generalization thanks to “engaging in acts of justification” and “attending to the quantitative relationships” of performing scalar arithmetic on the magic squares (Ellis, 2007). While I did not know of this bijective relationship between justification and generalization at the time, it helped make sense of why the

students went towards generalizing scalar operations with a variable “like n or x ” to make a magic square with a magic sum of “ $15 + 3n$ ”. In fact, they quickly generalized the operation to the point of recognizing it not only worked with integers, but also irrationals (π) and imaginary numbers. As such, their propensity to discuss, test, and justify their conjectures helped not only with their progress and both individual and group reasoning, but it also helped to establish a better basis for effective generalization.

It is worth noting again that these students are part of a friend group, and are thus more likely to engage in conversation, as well as be easily distracted, such as is the case with Student D. In fact, Student D appeared to be on vacation with friends, which actively contributed to his distracted behavior. Despite this particular student’s setbacks, with the other three students pooling their ideas and patterns together, they succeeded in the “effective coordination of a joint problem-solving space” (Barron, 2003); both the “content space” of the problem and “relational space” of discussion were utilized properly in their endeavor.

Factors Contributing to Afternoon Pair Progress

Unlike the morning group, the pair did not know each other as well, given they were from different graduating classes and likely had not interacted much with each other. I hoped that their more advanced coursework could counterbalance this faulty pairing dynamic; unfortunately, that was not the case. This disjoint behavior contributed to the breakdown of the previously cited “coordination of a joint problem-solving space”: although they cooperated in their “content space”, the “relational space” could not be properly built in a way to facilitate active discussion and cooperation necessary to solve the problem with an open mind.

Referencing these two students’ ratings of each other based on the following exit ticket questions, “On a scale of 1 to 10, how often do you think your comments/ideas/strategies were considered or discussed by/with your group members?” and “On a scale of 1 to 10, how much effort would you say you put into the group activities compared to the rest of the group?” – 1 was the lowest possible rating, 10 was the highest possible rating – Student K gave the ratings 9 and 3, respectively, while Student J gave the ratings 4 and 6, respectively. These ratings make sense: Student J had quickly discovered two of the patterns that generated magic squares, and, despite explaining it to Student K, Student K mostly ignored it and continued using his own method of initially defining the diagonal and structuring the rest of the square from it. These ratings and disjointed nature point to a failure in “establishing joint attention” (Barron,

2003). During the initial magic square problem, Student K also seemed confused by some of Student J's methods, namely the "donut" method. I had not anticipated the donut method when formulating the lesson and study, but it has similarities to the constraining method – placing one of the numbers first and then trying to solve the rest of the square – without actually constraining anything; rather, they effectively removed half of the relevant equations and failed to make meaningful progress: Student K said, "I was confused by creating a 'donut', and we proved that the method did not yield many results by trial and error."

Worthy of note is their opposing responses to the Exit Ticket question, "Before your peer explained their method, how were you visualizing or approaching the magic square? How did your understanding change or expand after hearing their explanation?" Student K noted how "interaction with my peer did not meaningfully affect my approach to the problem"; Student J noted, "I was attempting to create the magic squares by making the square a 3x3 matrix full of a single value and altering the values to try to maintain the properties of the matrix. I stopped because Student K discovered the one possible square. I think my method could've eventually produced the magic square though — it might've been a while though."

Thoughts on Lesson Plan Execution

I underestimated how long it would take students to solve the magic square. Initially, I wanted students to work individually on the problem for 3 minutes and share their ideas and strategies with group members for 7 more minutes. Instead, I gave them 6 minutes individually and another 9 minutes together. However, I expected the ease or difficulty in applying prior knowledge and finding patterns. I expected that both groups would figure out the magic sum using the sum formula; I knew that trial and error and Student B's pursuit of a valid numerical sequence would be difficult; I knew that even if Student K used a knight method, he would neither intuitively wrap his moves around the square nor use a break move.

Being strapped for time because of that, but noting we only had one group, I effectively had to shorten the Explain section to 10 minutes. Regardless, there was no need for them to explain their ideas to other students; they were the only students for the morning. The explanations were effectively intertwined with the Explore section, as students made conjectures, tested them, and explained what worked and what didn't work. However, given that the research targeted individual and group reasoning, the Elaborate section was not particularly needed for data. I could

have executed it better, and I could have accounted for the technical mishaps and mismatched data types as well.

For the afternoon section, I let the duo take more time, and it was clear that the magic square problem was not necessarily easier to solve because of more mathematical knowledge. I also let them answer the questions in the Google Slides; however, I quickly realized it occupied too much time (10 minutes) for them to fill out a few slides. In fact, I realized students could not fill out the Google Slides while also focusing on finding magic squares and their patterns. In hindsight, I could have made it more explicit, but the sheer number of applications I wanted them to use was clearly overwhelming, even for me on two monitors. Additionally, I needed to resolve instances of lack of collaboration, whether by strictly requiring groups of 3 or 4, or by better preparing for a participant not fully considering other students' ideas.

Across both sessions, I kept the Elaborate section short, as it was not the target of the research. However, I must address the issues with the section, notably that I had planned to use student examples for displaying the axioms, but could not feasibly implement it due to not only a lack of data on real-time student magic squares, but also the insurmountable issue of the time required to code the matrices into the TeX file itself. Additionally, the time constraints made it infeasible to discuss how to prove whether a set was a vector space or not, rendering my Exit Ticket application questions on vector spaces useless for student evaluation. While their responses to those questions were not used in this study, the failures with this section's execution must be considered.

As such, I would make the following changes:

1. Increase the amount of time for the Explore section by 5 minutes.
2. Preemptively create matrices for demonstrating the ten vector space axioms in the Elaborate section.
3. Strictly enforce groups of 3-4 with no exceptions.
4. Reformat the Google Slides to be less time-intensive or more useful, or replace it with a simpler Google Doc that would be more practical.
5. Expand the lesson to span more than one day to ensure that students are not forced to learn and understand material in a hyper-compressed time period that would hinder their learning and understanding instead of helping them.
6. Add additional questions and safeguards to ensure all students meaningfully participate and collaborate.

Conclusion

The study's findings demonstrate that students more effectively take up, discuss, and revise their conjectures when they actively collaborate and share their ideas and findings. Students in the morning group kept an open mind and used co-construction efficiently by building off each other's ideas, reasoning, challenging, and justifying these ideas, leading to a correlated generalization and synthesis of their ideas (Ellis, 2007). Conversely, the afternoon pair experienced noticeably lower levels of social and content cooperation, which, in turn, did not result in any effective taking up and revising of ideas, the lack of which was noticeably coming from Student K, who took no note of Student J's quick discoveries of two important transformations and continued using his own diagonal-first, fill later method. As such, they were "on-task but not engrossed" (Mulryan, 1992), but the lack of active cooperation and co-construction heavily counterbalanced their higher mathematical maturity. The afternoon pair may have lacked collaboration due to the absence of a third member to provide additional input or stimulate their ideas; However, Student K's non-collaborationist participation would have resulted in the same diminished quality of discussion.

Implications

This stark contrast reveals that mathematical knowledge is not necessarily as important, but it is secondary in comparison to effective group cooperation and sharing of concepts and ideas. Teachers should ensure that students are productively cooperating – like the morning group – and not working in silence – like the afternoon group. While assigning student roles was not used in this study, it helps to support and facilitate group discussion and revision of conjectures (Allen, 2012). I also suggest that teachers notice all students' ideas and exercise responsive teaching to reiterate students' ideas, fueling student thinking and the creation of ideas. By ensuring that students almost always are using each other's ideas and conjectures, they are more likely to develop and improve their own ideas in the process, facilitating co-construction for optimal group collaboration and avoiding potential instances of skip-connecting. These ideas can also coalesce into one big generalization that can be improved by making separate mini-conjectures, reinforcing and sharpening the overall ideas of the group.

It is also worth noting that this study was conducted in an online setting, which led to a dependency on online resources and handouts. Given physical screen limitations, these resources could not be adequately utilized without causing either more time constraints or attention overload, which would destroy active co-construction. As such, if

teachers must use online resources and handouts, they are advised to minimize the number of handouts or consolidate them into one file, such as a Google Doc. If conducted in an in-person environment, paper could and should be used over digital assets, with copies of questions and magic square templates being provided for each student, and an extra copy so that each group may collectively consider the provided questions.

Additionally, when it comes to off-task behavior, teachers should aim to get the student back on task to, at minimum, “minimally on-task” (Mulryan, 1992); however, if getting such a student back on task continuously fails or is rendered impractical, especially after 2 or 3 attempts, it is best to remove the student from the group to ensure that other members of the group may stay on-task and not be distracted; a direct talking with the student about the implications of their behavior is likely necessary if they are intentionally distracting the group. In my study, the not-on-task student, Student D, disconnected before his non-participation escalated, but it would be best practice to follow the above suggestion to ensure everyone else can participate without distraction.

If the lesson used in this study is used in a real classroom setting, teachers are recommended to use the lesson up to the Explain section as a starting point for a linear algebra class in general; it is better used as an overall base for introducing many linear algebra concepts, including, but not limited to:

1. Solving systems of equations using matrices, and defining whether the system is inconsistent, consistent but not unique, or unique. This involves using the equations defining the rows, columns, and diagonals. Solving also involves the instruction of (reduced) row echelon form, representing the solution as a parameterized form, and determining homogeneity.
2. Spans, linear combinations, and linear independence or dependence. This involves using scalar arithmetic operations students found during the lesson.
3. Linear and matrix transformations, vector spaces and subspaces, basis, and dimension. Students would be unknowingly performing linear and matrix transformations during the lesson, and the transformations they find correlate to some vector space axioms used to define a vector space of 3-by-3 magic squares, which is a subspace of all 3-by-3 matrices. Additionally, the 3-by-3 magic square vector space can be represented as the span of three

base matrices using the variables in Figure ?? . This magic squares vector space is equivalent to

$$Span\left\{\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \begin{bmatrix} -1 & 1 & 0 \\ 1 & 0 & -1 \\ 0 & -1 & 1 \end{bmatrix}, \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}\right\}$$

4. Eigenvalues and eigenvectors. Notice that each magic square, represented as a matrix with magic sum s , has an eigenvector $\langle 1, 1, \dots, 1 \rangle$ with corresponding eigenvalue s .

This recommendation will avoid the extreme time constraints that plague this lesson and provide maximal activity usage as a stepping stone for instruction of these linear algebra concepts.

Reflection

Possible Ways to Fix Execution

The following improvements are sorted by their relevant sections.

Recruitment Mishaps

Some improvements that could help to alleviate the problems with participant recruiting are as follows:

1. Not using unaffiliated secondary emails as a way of collecting the emails of interested participants.
2. Pursuing more engaging or enticing advertisements that could better draw interested people's attention.
3. Pursuing more direct forms of communication with people using direct or private messages.
4. Ensure the Participation Interest Google Form fully supports a group-only section for those filling out the form.

Also include an explicit warning that multiple sections exist, as people may click "Next" and think that they submitted the form.

Meeting Chaos

The chaotic nature of participant arrival and the meeting itself exposed several improvements that I needed to make if I wanted to implement this again:

1. Add a question to the Participation Interest form about whether they would join using a computer or phone, with explicit recommendations to join via a computer for an optimal experience.
2. Be more explicit about arriving on time.
3. Emphasize that if participants are unable or unwilling to participate and be on time, or if there are any possible issues with work priorities, then they should let me know beforehand.
4. Secure the meeting with a password and explicitly state in puzzle instructions that they may not receive any outside help, while also encouraging them to ask each other or the teacher.

Technical Mishaps

To ensure I don't run into technological issues as discussed earlier, I would need to learn and better understand LaTeX formats, understand where errors commonly occur, and how to fix them. I should also consistently check the PDF to ensure that everything is updating properly. These two improvements would ensure that the lesson is not caught in major technical issues that take more than a minute to fix.

Improvements Based on Student Thinking

Given that the generalized magic squares problem proved to be a long, albeit intriguing mathematical inquiry into magic square transformations and their corresponding vector space axioms, I believe that creating some parallel to basic arithmetic, i.e. suggesting that students consider "transforming" the magic sum and copying it back to its corresponding magic square, may be beneficial in using prior student knowledge, while also not causing extreme numerical chaos prone to arithmetic mistakes. The mechanism behind this concept is as follows: for M as the domain that is the vector space of 3-by-3 magic squares, and $\mathbb{R}$ as the codomain, we can make an onto function $f : M \rightarrow \mathbb{R}$ which maps a magic square to the value of the middle square. For a coordinate pair (m, c) for $m \in M$ and its corresponding middle value $c \in \mathbb{R}$, we can have students consider a particular coordinate pair, consider operations

and all scalar arithmetic properties on c , and then consider that same operation on all entries of its corresponding m . While this concept is imperfect, as it is not a bijection, creating coordinate pairs helps to radically simplify this concept to align more closely with students' prior experience and make the connection to vector space axioms more understandable without much complexity.

Personal Reflection

Despite several issues that arose in the execution of the lesson plan, research, and data collection, I feel much more prepared for the professional aspects of being a teacher. Although nothing was perfect, I know what ideas and concepts I devised worked, what issues to fix or remove entirely, and how to improve my overall understanding of teaching in a professional sense, all of which I have already explained in great length. In the process, I was able to test teaching ideas like using Visual Studio Code for both live teaching without the need for writing anything and field notes, such that it's as interactive as possible and student-led like a whiteboard/blackboard style lesson, but without the disadvantage of a slow thinking-to-writing time, bad handwriting, or the rigidity of traditional digital lesson methods, i.e. Google Slides or a PowerPoint.

Additionally, I feel that this instance has been a much better indicator of what I can and should improve than previous small teaching or tutoring instances I have undertaken. I better understand what technological errors could occur, and how to fix them. I also know how to manage time better or work around it to prevent shortening parts of the lesson. Importantly, I feel much more motivated to use non-routine problems in future lessons, as they are not only rich in both content and usage, but also great for eliciting student thinking and collaboration, skills that are important for success in school, college, and beyond.

However, I still have questions about teaching, specifically about employing non-routine problems for classroom learning:

1. Can every mathematical concept be discussed using a non-routine problem?
2. How can teachers employ and adapt these non-routine problems without dealing with curriculum or timing problems?
3. Why do most teachers not employ these problems? Is it because of teaching habits?

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Appendix

The appendix consists of comments on transcript availability and Exit Ticket self-reflection responses used, divided into the morning and afternoon sessions. Again, please note that exit ticket responses have several typos and errors made by the participants, and obvious fixes were made in accordance with APA guidelines while preserving the participants' original meanings.

Morning Section

Transcript

Please note that all relevant portions of the morning group's transcript were already included in the paper itself and are effectively consecutive; no other portions of the transcript are of note.

Morning Group Exit Ticket Self-Reflection Responses

For reference, here are the following questions:

1. On a scale of 1 to 10, how successful do you think your group was in finding other magic squares?
2. On a scale of 1 to 10, how often do you think your comments/ideas/strategies were considered or discussed by/with your group members?
3. On a scale of 1 to 10, how often did you feel that someone was not contributing to your group?
4. On a scale of 1 to 10, how much effort would you say you put into the group activities compared to the rest of the group?
5. When you were first asked to create new magic squares, what was the first idea or mathematical operation you tried? Did it work as expected?
6. What patterns did you personally discover? How did you discover it?
7. Are there any patterns that you thought could create a magic square, but didn't quite work? How did you adjust your approach after that?
8. Describe a moment during breakout rooms when a group member explained their strategy or pattern discovery. What was their idea, and how did it differ from your initial way of thinking about the problem?
9. Before your peer explained their method, how were you visualizing or approaching the magic square? How did your understanding change or expand after hearing their explanation?
10. Was there a pattern suggested by a peer that you initially disagreed with or were confused by? How did your group test or prove whether it worked?
11. How did seeing multiple valid patterns (like adding two magic squares or multiplying by a scalar) change how you think about "magic squares" as a broader mathematical system rather than just isolated puzzles?

Responses by Student A:

1. 7
2. 7
3. 8

4. 5
5. At first, I tried adding the same number to every square in the larger magic square. It worked because it followed the established axioms (although I didn't know the axioms at that time).
6. I personally discovered that multiplication and addition of any number from our original magic square kept the correct form going forward.
7. No, not really.
8. Student B explained his thought process by looking at the grand scheme of the set, finding the middle value and putting that in the center. It changed my initial way of thinking.
9. I was simply just doing guess and check, but Student B's advice helped me see the bigger picture.
10. I was initially confused with Student C's idea about multiplying the squares by each other, but it ended up working out. He showed me an example to explain it to me.
11. Yes, I think it did.

Responses by Student B:

1. 7
2. 10
3. 6
4. 8
5. At first, I tried addition, and it worked as expected because it only changed the axis of the vector space but not the properties.
6. I personally discovered that once you had a base vector space, you could modify the numbers with any simple operation, (addition, subtraction, multiplication, division), as long as you did so to all numbers in the space when I was testing out the different operations on the base set of 1-9.
7. I tried to use multiples of numbers, but I realized that they did not have uniform pattern in correspondence to the rest of the numbers in the vector space. Instead, I decided to use duplicates that would have a mirrored version on the other side of the middle number of my set of variables in the vector space.
8. When Student A showed that you could add any given number to an already existing vector space to get a new one it gave me the idea to try other operations and see if it would work with non integers.
9. I was initially visualizing/approaching the magic square as a linear set of numbers. However, after hearing my peers explain their method, my understanding changed because I realized that the set of numbers did not need to be uniform and expanding in an increasing order but could take on many forms as long as they met the criteria of the ten axioms.
10. At first we thought that any group of random real numbers could work as a vector space so we attempted to create a magic square using the numbers but found it to be impossible.
11. Seeing multiple valid patterns changed how I saw "magic squares" because it made me recognize them as spaces that followed actual arithmetic properties and not just a puzzle with numbers.

Responses by Student C:

1. 7
2. 7
3. 5
4. 7
5. I tried adding all the numbers 1-9 and it worked but I didn't know where to go with the info.
6. The patterns of addition, subtraction, multiplication, division by a constant create a new matrix.
7. Not using uniform magic squares doesn't work.
8. Student B mentioned his way of thinking, and it differed from mine because he opened my eyes to how to understand and break down the way the sum of 15 is the magic number.

9. Initially, I approached it by guess and check, but after, my opinion changed by using a better stepped approach.
10. Not really.
11. It gave me a broader perspective and made me understand that there are so many possibilities to magic squares.

Student D failed to respond.

Responses by Student E:

1. 8
2. 7
3. 6
4. 9
5. Just trial and error. It didn't work though.
6. I discovered that the pattern was similar to others.
7. I tried making the rows add up correctly first, but I realized the columns and diagonals no longer had the same sum. Instead, I adjusted the numbers together so changing one value would still keep the row, column, and diagonal sums balanced.
8. One of my group members mentioned looking at the relationships between opposite positions in the square instead of focusing on each row individually. It differed from my initial thinking because I was only checking whether each row added to the correct total, while they were looking for a pattern that kept the whole square balanced.
9. They helped me a lot with the puzzles.
10. No there wasn't.
11. It didn't really change much. Just a bunch of trial and error.

Afternoon Section

Transcript

I provided as much of the afternoon group's transcript that I deemed usable; the rest is not particularly helpful to include.

Afternoon Group Exit Ticket Self-Reflection Responses

Questions provided were the same as for the morning section. Please reference Morning Group Exit Ticket Self-Reflection Responses for the questions if necessary. Responses by Student J:

1. 7
2. 4
3. 4
4. 6
5. At first I tried to do a guess and check, but this was just to gain an intuition for the problem and I didn't expect it to work.
6. I saw that the magic squares could be translated using basic arithmetic operations.
7. I thought that maybe the primes might have a contribution because I noticed in some of the squares Student K made, there were a lot of primes, but I didn't pursue this idea long enough to get anywhere.

8. Student K showed me how he placed the highest value on the side instead of the corner and it made me reconsider how I was thinking about creating the squares.
9. I was attempting to create the magic squares by making the square a 3x3 matrix full of a single value and altering the values to try to maintain the properties of the matrix, I stopped because Student K discovered the one possible square. I think my method could've eventually produced the magic square though — it might've been a while though.
10. We initially were doing a lot of knights moves, but I gave up on that because it didn't seem like we were getting anywhere. Student K returned to this and found a correct square so it turned out to be a valuable strategy.
11. It made me think of it as a vector immediately and I started thinking about the most interesting ways to manipulate it.

Responses by Student K:

1. 8
2. 9
3. 2
4. 2
5. I initially started with the idea of having all the numbers opposite each other add up to the same number and having 5 in the middle because it is the mean of numbers 1-9, I couldn't get that to work because I put a 9 in the corner, so I moved on.
6. That magic squares can be added together and result in a magic square when I was experimenting with the magic squares I had.
7. I tried making the sum 16 but then I realized that was not possible so i decided to stick with 15.
8. Student J suggested creating a “donut” that works by first ignoring the center of the magic square.
9. Interaction with my peer did not meaningfully affect my approach to the problem.
10. I was confused by creating a “donut” and we proved that the method did not yield many results by trial and error.
11. It helped me understand the relationship between magic squares and vector spaces.